

Learning Manipulation-Sufficient Representations via Outcome Bottlenecks

Md Selim Sarowar and Sungho Kim*

Abstract: Networked manipulation endpoints couple perception to actuation across compute- and bandwidth-limited links, yet commonly exchange dense geometric states optimized for fidelity rather than action outcomes. A stochastic representation is learned with a policy-free, action-conditioned outcome bottleneck: marginal outcome log-loss supplies distortion and a KL term regularizes rate. The construction is motivated by the minimal statistic that preserves the outcome distribution of every admissible action, while the implemented finite model is evaluated as a rate-regularized mixture predictor. The same encoder and outcome head support grasp selection, singleton conformal filtering, active viewpoint selection, and latent test-time adaptation. A finite-probe theorem identifies the local level-set tangent space with the null space of an outcome Jacobian. The synthetic oracle verifies this result; on scanned objects, an analytic surrogate agrees with measured simulator invariances within 0.56° . Across 11,979 simulated grasps on 13 objects, a reconstructed-geometry wrench score attains 0.542 AUC against lift success and falls below chance on curved objects, while our representation attains 0.876. At 25% commitment, executed-grasp success is 0.503 versus 0.984. The 512-byte interface is $288\times$ smaller than one RGB-D frame and runs at 16 ms per CPU decision. Independent synthetic points track the tested conformal levels; scanned-object all-pair coverage is reported as a clustered empirical diagnostic. On unseen objects, within-scene AUC falls to 0.569, and a full-feedback update raises empirical mean pairwise coverage from 0.728 to 0.883.

Index Terms: Edge intelligence, information bottleneck, robotic grasping, conformal prediction, active perception, Internet of Things, semantic communication.

I. INTRODUCTION

ROBOTIC manipulators are becoming Internet of Things endpoints. Warehouse picking cells, mobile manipulators on factory floors, and service robots in shared spaces increasingly run their perception either on embedded processors with hard compute ceilings or offloaded across a wireless link to an edge server with hard bandwidth ceilings [1]–[3]. In both regimes the systemic bottleneck is the same object: the message that perception hands to decision making. Current grasping pipelines make that message a dense geometric state. A reconstruct-then-plan system transmits or stores a 6D pose together with a completed mesh or truncated signed distance field [4], [5]; a feature-field system attaches high-dimensional descriptors to every point of a radiance or Gaussian representation [6], [7]; an end-to-end regressor consumes the full point cloud on every invocation [8]–[10]. A single 96×96 RGB-D frame already carries 36,864 scalars; the multi-view sets these

pipelines prefer carry an order of magnitude more, and meshes and fields more still.

The cost would be acceptable if fidelity of the transmitted state predicted the success of the action executed on it. It does not. On 13 scanned objects (Section VIII) the Ferrari–Canny wrench-space quality [11], computed on an oriented bounding box of the kind a pose-and-shape pipeline recovers, scores 0.542 AUC against a rigid-body lift of the true mesh, barely above chance, and falls *below* chance to 0.473 on curved objects: the grasps it prefers are the grasps that fail. The failure diagnoses this evaluated reconstruction–metric pair: the reconstruction is self-consistent but wrong at contact locations, so its analytic score inherits the error [12], [13].

Perception for manipulation is formulated as estimation of a *manipulation-sufficient statistic*: the minimal code z such that the conditional outcome distribution $p(y|o, a)$ of every admissible action a is preserved when the observation o is replaced by z . The definition induces an equivalence on world states, two states being indistinguishable when every action yields the same outcome distribution, and perception becomes estimation of a stochastic representation of the quotient. Three consequences follow. First, an explicit KL rate surrogate encourages a compact representation and is measured in nats, which makes it a compact interface for a bandwidth-limited perception-to-decision link and aligns the formulation with goal-oriented communication [14], [15]. Second, the stochastic encoder exposes predictive spread, so pairwise conformal filtering under exchangeability, active acquisition of the next viewpoint, and assimilation of executed-probe outcomes all become inference procedures over the same two learned modules rather than additional subsystems. Third, the formulation admits a finite-probe local theory: the null space of an outcome Jacobian identifies infinitesimal state directions invisible to the chosen probe family, and the diagnostic is verified numerically.

The contributions are:

- 1) **A policy-free, action-conditioned outcome bottleneck** (Section IV) whose distortion is the negative log-likelihood of task outcomes over the admissible action family rather than a reconstruction, pose, or reward term. An idealized form of the objective recovers the minimal sufficient partition of the observation (Theorem 3); the trained system optimizes the rate-regularized marginal log-loss of a mixture predictor, which Remark 1 distinguishes from that ideal; Proposition 1 identifies the marginal-likelihood estimator (Proposition 1) without which the objective collapses, together with a per-outcome-dimension budget that task-conditions the representation.

- 2) **An identifiability theory for outcome-defined perception** (Section V). For a fixed finite probe family, the local level set through x is a submanifold whose tangent space is the null space of the finite-probe outcome Jacobian $J_N(x)$ (Theorem 5); pose readouts are locally constrained only on row $J_N(x)$, and full-state metrics can penalize directions that the probe family cannot distinguish (Corollary 1). The predicted null spaces of an analytic surrogate match independently measured simulator outcome-invariant subspaces to within 0.56° of principal angle on the rank-deficient scanned objects; the theorem itself is verified on the synthetic oracle where the differentiated map is the true kernel.
- 3) **A shared outcome model for sensing and latent adaptation** (Section VI-C): viewpoint selection maximizes the expected reduction of an outcome-space ambiguity $U(o)$, while probe assimilation uses the same learned head in a distinct constrained variational update whose unrestricted fixed point is characterized in Proposition 4.
- 4) **A fail-closed pairwise conformal filter** (Section VI-B): for exchangeable scene–action units, a split-conformal construction over the mixture success probability retains an action only when its prediction set is the singleton $\{1\}$ and abstains otherwise. That theorem is separated from the clustered all-pair diagnostic on the scanned corpus and evaluate an anchored full-feedback level update under simulated drift without claiming selective-feedback validity.

Everything is validated in a two-tier physics stack in which the *analytic* tier plans on reconstructed geometry and the *simulation* tier executes on the true convex decomposition, so the gap between reconstruction and reality is measurable inside simulation with no robot in the loop (Section VII). The proposed system, labeled *Ours* in every table, is compared against the analytic wrench-space pipeline, random and majority-class selection, and a privileged-state baseline fit on the 14-vector (1). The scope is limited to data generated by the simulator/analytic stack; the representation distribution is a diagonal Gaussian, the full model is trained with three seeds and each ablation and leave-one-object-out configuration with one; Section IX enumerates what these choices do and do not license.

II. RELATED WORK

A. Reconstruct-Then-Plan Pipelines

The dominant architecture estimates a geometric state and hands it to a planner: pose estimators [4], [5], [16] regress an SE(3) transform, completion densifies occluded surfaces, and grasp planners evaluate force closure or the Ferrari–Canny radius on the recovered geometry [11], [12]. It inherits two structural limitations: the estimand is ill-posed, a canonical frame not existing for symmetric or partial objects, and geometry-wide metrics such as ADD-S and Chamfer [17] need not weight errors according to their effect on the evaluated grasp outcomes. Theorem 5 gives a finite-probe local diagnostic for the first issue, and Section VIII measures the second for our baseline: analytic quality on reconstructed geometry

is uninformative about realized lifts, worse than chance where the reconstruction is wrong.

B. End-to-End Grasp Synthesis

Learned grasp regressors map observations to grasp poses and scalar scores [8]–[10], [13], [18]. They are implicitly outcome supervised but their perceptual state is typically not exposed as a stochastic interface for joint selection, conformal filtering, viewpoint valuation, and probe assimilation. Their reported scalar scores are not, by default, a calibration guarantee for the outcome and action family studied here. The proposed method retains outcome supervision while exposing a compact, transmissible stochastic representation from which those four procedures derive.

C. Task-Oriented Representations and Information Bottlenecks

The information bottleneck [19], [20] formalizes compression subject to relevance and has been applied to control, most directly by Pacelli and Majumdar [21], who compress state toward the reward of a single fixed policy. Feature fields and affordance methods [6], [7], [22] attach task-relevant descriptors to geometry but keep geometry as the substrate, and vision-language-action models [23] absorb perception into a monolithic policy. Our estimand differs in kind from all three lines: sufficiency is defined with respect to the *entire action-conditioned outcome family* $\{M(\cdot | x, a)\}_a$, policy-free and embodiment-parameterized, the object of estimation is a quotient of the state space with an explicit identifiability limit (Theorem 5), and the output supports a conformal set-valued decision rather than only a feature vector. The distinction is the combination of an action-family estimand, a finite-probe local diagnostic, and a pairwise conformal construction, rather than a claim that the cited systems cannot support uncertainty handling in other forms.

D. Calibrated Uncertainty and Abstention in Robotics

Conformal prediction [24], [25] supplies finite-sample, distribution-free coverage and has entered robotics through planner-level constructions [26] and language-model task planning [27]. Epistemic-uncertainty estimation via ensembles or variational posteriors [28] supplies the raw signal. Our construction places the conformal filter at the perception interface: the nonconformity score is computed on the mixture success probability, the filtered object is an *action set*, and the empty set is a typed abstention rather than a fallback. The finite-sample theorem applies to exchangeable scene–action units; under drift, a separate full-feedback adaptive-level experiment is reported [29].

E. Goal-Oriented Communication and Edge Offloading

Semantic and task-oriented communication argues that networked systems should transmit what the receiver needs for its task rather than reconstruct the source [14], [15]; split computing studies where to cut a network across the device-edge boundary [30]. What this literature lacks for manipulation

is a definition of *which* bits the task needs. The manipulation-sufficiency objective supplies it: the KL term upper-bounds the representation's mutual information and can motivate a code only when an appropriate coder is implemented [31], [32], the distortion is the task loss, and the operational interface is a 128-scalar parameter vector. This is a concrete instance of goal-oriented compression whose goal is physical and whose sufficiency claim is testable against executed outcomes.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Deployment Architecture

The system uses the standard two-node edge topology [1], [30] (Fig. 1): a *sensor node* runs the stochastic encoder, and a *decision node* runs the outcome head, calibrator, and action selection. The only message that crosses the interface per decision is the encoder-parameter pair $(\mu, \log \sigma^2) \in \mathbb{R}^{2d}$ with $d = 64$, i.e., 128 scalars against 36,864 for one 96×96 RGB-D frame and 294,912 for the eight-view set used by our active-perception experiments, reductions of $288\times$ and $2304\times$ respectively. The training objective also reports a KL rate surrogate that upper-bounds $I(Z; O)$; at the primary operating point it is 0.534 nats. This is not a measured packet length: no quantizer or relative-entropy coder is implemented [31], [32]. All results concern the statistical properties of the 512-byte fp32 interface; no radio link is simulated, and no protocol-level latency is claimed.

B. Spaces, Kernels, and the Generative Model

The world state $x = (S, T, \phi) \in \mathcal{X}$ comprises object shape S , pose $T \in \text{SE}(3)$, and physical parameters $\phi = (\mu_f, m, c)$: friction coefficient $\mu_f > 0$, mass $m > 0$, and center of mass $c \in \mathbb{R}^3$. For each fixed object mesh, the continuous rigid-object state is a smooth $d_{\mathcal{X}}$ -dimensional manifold; the scanned-object identifiability calculation is conditional on that mesh and uses

$$x = (t, r, \log e, \log \mu_f, \log m, c) \in \mathbb{R}^{14}, \quad (1)$$

with translation $t \in \mathbb{R}^3$, a local axis-angle coordinate $r \in \{u \in \mathbb{R}^3 : \|u\| < \pi\}$ on a fixed logarithm branch, log half-extents $\log e \in \mathbb{R}^3$ of the fitted bounding box, and the physical parameters. The discrete mesh identity remains part of S outside this conditional 14-vector. A scene prior $p(x)$ is induced by dropping objects onto a plane and letting them settle under gravity. The observation space is $\mathcal{O} = \mathbb{R}^{H \times W \times 4}$ (RGB-D), extended to view sets $\mathcal{O}^V$ for multi-camera acquisition, with sensor kernel $p(o|x)$ realized by a calibrated renderer. The action space is

$$\mathcal{A} = \text{SE}(3) \times \mathbb{R}_{\geq 0}, \quad a = (t_a, r_a, w) \in \mathbb{R}^7, \quad (2)$$

a grasp pose and gripper opening width. The rotation coordinate r_a uses the same fixed local logarithm branch as r in (1). A target reference measure ρ is supported on the admissible set; a gripper descriptor $g \in \mathcal{G}$ conditions the outcome head so that sufficiency is defined with respect to a gripper family rather than one embodiment. The outcome space is

$$\mathcal{Y} = \{0, 1\} \times \mathbb{R} \times \mathbb{R}_{\geq 0}, \quad y = (\text{succ}, \text{margin}, \text{slip}), \quad (3)$$

TABLE I
PRINCIPAL NOTATION. SYMBOL COLLISIONS WITH COMMON SINGLE LETTERS ARE AVOIDED: THE LATENT PRIOR IS $p_0(z)$, BOX HALF-EXTENTS ARE e , AND THE SENSING BUDGET IS B_{sense} .

Symbol	Meaning
$x \in \mathcal{X}$	continuous state (1), $d_{\mathcal{X}} = 14$ per fixed mesh
$o \in \mathcal{O}$	RGB-D observation (or view set)
$a \in \mathcal{A}$	grasp action (2), $a \in \mathbb{R}^7$
$b \in \mathcal{B}$	sensing action (viewpoint index)
$y \in \mathcal{Y}$	outcome (succ, margin, slip)
$M(\cdot x, a)$	task-outcome kernel used by the theory
ρ	target reference measure on admissible actions
ν	boundary-focused training proposal
$q_\theta(z o)$	stochastic encoder, $z \in \mathbb{R}^d$, $d = 64$
$p_\psi(y z, a)$	outcome head
$\bar{p}_\psi(y o, a)$	mixture predictor $\mathbb{E}_q[p_\psi(y z, a)]$
$p_0(z)$	latent prior $\mathcal{N}(0, I_d)$
β	per-outcome budget
$(\beta_{\text{succ}}, \beta_{\text{m}}, \beta_{\text{s}})$	
$s(o, a), v(o, a)$	mixture success mean/variance
$U(o)$	outcome-space ambiguity (29)
$\hat{q}$	conformal quantile; α : miscoverage target
$J_N(x)$	finite-probe outcome Jacobian (19)

binary lift success, a wrench-space stability margin, and a post-lift slip magnitude. Let $M : \mathcal{X} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{Y})$ denote the task-outcome kernel. It is generic in the theory; Section VII-A separates its synthetic instantiation from the scanned-corpus simulator outcomes and analytic margin label. Here $p(y|o, a)$ denotes the interventional predictive $p(y|o, \text{do}(A = a))$. Under the target action measure, one datum is generated as

$$\begin{aligned} x &\sim p(x), & o &\sim p(o|x), & a &\sim \rho, \\ y &\sim M(\cdot|x, a), & z &\sim q_\theta(\cdot|o). \end{aligned} \quad (4)$$

where $q_\theta(z|o)$ is the stochastic encoder distribution over the latent space $\mathcal{Z} = \mathbb{R}^d$, and $p_\psi(y|z, a)$ is the learned outcome head. For efficient corpus construction, training may instead draw $a \sim \nu(\cdot|x)$ from a boundary-focused proposal. Assume $\rho \ll \nu(\cdot|x)$ and use weights $w(x, a) = d\rho/d\nu(\cdot|x)$; the implemented self-normalized estimate is a biased but consistent approximation to the target ρ -risk. The target measure ρ and proposal ν are therefore distinct. The target action draw is independent of (X, O, Z) . Conditioned on $A = a$, the variables form the Markov chain

$$Z - O - X - Y, \quad (5)$$

and the data-processing inequality gives, for every a ,

$$I(Z; Y|A) \leq I(O; Y|A) \leq I(X; Y|A). \quad (6)$$

No statistic of the observation can carry more action-relevant outcome information than the observation itself. The formulation seeks the smallest Z that makes the left inequality in (6) tight. Table I collects the principal symbols.

C. The Estimand

For an observation o , define the posterior-predictive outcome field

$$\begin{aligned}\eta(o) &:= (a \mapsto p(y|o, a)), \\ p(y|o, a) &:= p(y|o, \text{do}(A = a)) \\ &= \int_{\mathcal{X}} M(y|x, a) p(x|o) dx.\end{aligned}\quad (7)$$

Because o does not determine x , the field $\eta(o)$ is the most that any perception system can extract for manipulation: it assigns to every action its Bayes-optimal outcome distribution given the evidence.

Definition 1 (Manipulation sufficiency): A representation kernel $q(z|o)$ is manipulation-sufficient iff $Y \perp O \mid (Z, A)$ under the joint law induced by (4); equivalently, for ρ -a.e. action a ,

$$p(y|o, z, a) = p(y|z, a) \quad \text{for almost every } (o, z).$$

Because $Z - O - Y$ conditional on A , this is equivalent to $I(Z; Y|A) = I(O; Y|A)$. For a deterministic statistic $Z = t(O)$, it reduces to $p(y|o, a) = p(y|t(o), a)$ almost surely.

Definition 2 (Minimality): A sufficient t^* is minimal iff for every sufficient t , t^* is a measurable function of t .

Definition 3 (Manipulation indistinguishability): For $x, x' \in \mathcal{X}$, write $x \sim x'$ iff $M(\cdot|x, a) = M(\cdot|x', a)$ for ρ -a.e. a . The relation is an equivalence; $\mathcal{X}/\sim$ is the quotient with projection π , and the outcome map

$$\Phi : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})^{\mathcal{A}}, \quad \Phi(x) = (a \mapsto M(\cdot|x, a)) \quad (8)$$

satisfies $x \sim x' \iff \Phi(x) = \Phi(x')$.

These definitions pin the estimand: perception should estimate $\eta(o)$, equivalently a coordinate on $\mathcal{X}/\sim$ weighted by the posterior $p(x|o)$, and never a metric pose. Section V makes the geometry of the quotient explicit.

D. Standing Assumptions

- A1** All kernels are measurable; densities exist w.r.t. the reference measures where written.
- A2** M is queryable at training time, and the family $\{p_\psi\}$ can represent the true predictive at the optimum (realizability).
- A3** ρ has full support on the admissible action set of every scene.
- A4** The outcome Jacobian (19) has locally constant rank near the states of interest.
- A5** The split-conformal theorem uses one scene–action unit per independent scene, exchangeable between calibration and test. Nonexchangeable drift is evaluated empirically and is not covered by that theorem.
- A6** On the identifiability domain, $\mathcal{X}$ is a fixed-dimensional smooth manifold and Φ is C^1 in x for each a (rigid objects).
- A7** For active perception, either a generative observation model or an amortizable information-gain estimator exists; perturbing probes have a known transition.
- A8** The success functional is nonconstant in a on a set of positive ρ -measure for almost every scene.

IV. THE MANIPULATION-SUFFICIENCY OBJECTIVE

A. Variational Bound

The estimand of Section III-C translates into a Lagrangian over the encoder and head: minimize rate subject to sufficiency,

$$\min_{q_\theta, p_\psi} I(Z; O) - \beta I(Z; Y|A), \quad \beta > 0. \quad (9)$$

Both mutual informations are intractable; standard bounds make them so [20]. For any prior $p_0(z)$,

$$I(Z; O) \leq \mathbb{E}_o[\text{KL}(q_\theta(z|o) \| p_0(z))] =: R(\theta), \quad (10)$$

with gap $\text{KL}(q_\theta(z) \| p_0(z)) \geq 0$, and since $I(Z; Y|A) = H(Y|A) - H(Y|Z, A)$ with $H(Y|Z, A) \leq -\mathbb{E}[\log p_\psi(y|z, a)]$ for any head,

$$I(Z; Y|A) \geq H(Y|A) - \underbrace{\mathbb{E}[-\log p_\psi(y|z, a)]}_{=: D(\theta, \psi)}. \quad (11)$$

$H(Y|A)$ is constant in (θ, ψ) . Substituting (10)–(11) into (9), dropping constants, and generalizing the multiplier to a per-outcome-dimension budget $\beta = (\beta_{\text{succ}}, \beta_{\text{m}}, \beta_{\text{s}})$ yields the training loss

$$\mathcal{L}(\theta, \psi) = \sum_{j \in \{\text{succ}, \text{m}, \text{s}\}} \beta_j D_j(\theta, \psi) + R(\theta), \quad (12)$$

where D_j is the negative log-likelihood of outcome coordinate j under the factorized head. Two properties of (12) carry the reformulation. First, distortion lives in outcome space: there is no reconstruction or pose target, and the dominant supervision is executed outcome. Two of the three outcome coordinates, success and slip, are read from the simulator rollout; the margin coordinate, on the scanned corpus, is the analytic Ferrari–Canny value on the bounding box, so a down-weighted geometry-derived label does enter at $\beta_{\text{m}} = 10^{-2}$ (Section VII-B, and the ablation of Section VIII-H turns precisely on this). The claim is therefore that supervision is outcome-dominated with a small analytic-margin regularizer, not that geometry is absent from the loss. Second, β multiplies the distortion: increasing β emphasizes predictive fit and decreasing it emphasizes compression. The exact constrained ideal is Theorem 3; the finite- β implemented objective is not claimed to attain sufficiency. The reversed convention $L = \sum_j D_j/\beta_j + R$ traces the same frontier as a set but inverts every semantic statement about the knob, and a per-dimension budget under it allocates capacity to the wrong outcomes. The multiplier direction therefore affects the interpretation of the objective.

The budget is also the formal entry point for task conditioning: an instruction that cares about slip and not about placement re-weights $(\beta_{\text{succ}}, \beta_{\text{m}}, \beta_{\text{s}})$, changing which distinctions the statistic must resolve. The three D_j are likelihoods on incommensurate scales (a Bernoulli cross-entropy of order 0.6 nats against continuous densities of order 4 nats), so a numerically uniform budget is not a semantically uniform one; Section VIII-H measures the collapse a uniform budget causes. Throughout, β written as a scalar denotes β_{succ} with $\beta_{\text{m}} = \beta_{\text{s}} = 10^{-2}$.

B. Gaussian Instantiation and the Energy Form

The encoder is a diagonal Gaussian $q_\theta(z|o) = \mathcal{N}(\mu_\theta(o), \text{diag } \sigma_\theta^2(o))$ with prior $p_0 = \mathcal{N}(0, I_d)$, reparameterized as $z = \mu_\theta(o) + \sigma_\theta(o) \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I_d)$ [33], giving the closed form

$$\text{KL}(q_\theta(z|o) \| p_0) = \frac{1}{2} \sum_{j=1}^d (\sigma_j^2 + \mu_j^2 - 1 - \log \sigma_j^2). \quad (13)$$

The head factorizes as

$$\begin{aligned} p_\psi(y|z, a) = & \text{Bern}(\text{succ}; \varsigma(f_\psi(z, a))) \\ & \cdot \mathcal{N}(\text{margin}; m_\psi, \tau_\psi^2) \\ & \cdot \text{ZILN}(\text{slip}; \pi_\psi, \nu_\psi, \varrho_\psi^2), \end{aligned} \quad (14)$$

with ς the logistic function and ZILN a zero-inflated log-normal: a held grasp slips exactly zero, so slip requires a Bernoulli occurrence component on top of a log-normal magnitude. For the ideal variational objective, define the per-sample energy

$$E(z; o, a, y) = -\log p_\psi(y|z, a) + \log \frac{q_\theta(z|o)}{p_0(z)}, \quad (15)$$

whose expectation is (12) up to the outcome-budget weights. The implemented marginal objective below is not the expectation of this energy: it replaces the outcome term by the log loss of the mixture predictor.

C. The Marginal-Likelihood Estimator

How the expectation over z is estimated decides whether the objective works at all. With K encoder samples the two candidate estimators of the success distortion are

$$\begin{aligned} \hat{D}^{\text{post}} &= \frac{1}{K} \sum_{k=1}^K -\log p_\psi(y|z_k, a), \\ \hat{D}^{\text{marg}} &= -\log \frac{1}{K} \sum_{k=1}^K p_\psi(y|z_k, a). \end{aligned} \quad (16)$$

Proposition 1 (The marginal estimator matches the decision rule): $\mathbb{E}[\hat{D}^{\text{post}}] \geq \mathbb{E}[\hat{D}^{\text{marg}}]$ by Jensen's inequality, and $\hat{D}^{\text{marg}}$ is a consistent estimator of the negative log marginal likelihood $-\log \mathbb{E}_{z \sim q} [p_\psi(y|z, a)]$, which is the log-loss of the quantity $s(o, a) = \mathbb{E}_z [\varsigma(f_\psi(z, a))]$ that the decision rule (24) and conformal filter (27) consume. The two estimators optimize different objectives, and the gap is largest when the encoder distribution is noise-dominated.

The log-average has the same multi-sample form used by importance-weighted objectives [34], but here the samples come directly from the encoder measure defining the mixture and no importance ratio appears. $\hat{D}^{\text{post}}$ penalizes errors for every draw of z . When the encoder spread dominates its mean (measured on our corpus: $|\mu| \approx 0.37$ against $\sigma \approx 0.71$), this can favor a head that is insensitive to z . The associated collapse of action response is measured in Section VIII-H; the observation is empirical, not a uniqueness claim about the optimizer. All experiments train with $\hat{D}^{\text{marg}}$, $K = 8$, and anneal the rate weight from 0 to 1 over 15 epochs to escape the posterior-collapse fixed point at which $\partial D / \partial \sigma \equiv 0$.

Implemented objective. The variational relevance bound (11) contains the cross-entropy $\mathbb{E}_q[-\log p_\psi]$ ($\hat{D}^{\text{post}}$); the reported models instead minimize the rate-regularized marginal log-loss of the mixture predictor $\bar{p}_\psi(y|o, a) = \mathbb{E}_{z \sim q} [p_\psi(y|z, a)]$ ($\hat{D}^{\text{marg}}$). These differ, so the constrained minimal-sufficiency result of Theorem 3 is a motivating ideal, not a property the trained system provably attains (Remark 1). In particular, Theorem 1 applies only to a truly sufficient representation and is not inherited by $\bar{p}_\psi$ without an additional approximation guarantee. The method is therefore a rate-regularized mixture predictor whose bottleneck derivation motivates the estimand, and avoid claiming any reported code is a proven minimal sufficient statistic.

D. Sufficiency Guarantees

Theorem 1 (Sufficiency preserves attainable success): Let Z be manipulation-sufficient and define the true success functionals $s_O^*(o, a) = \mathbb{E}[\text{succ}|O = o, A = a]$ and $s_Z^*(z, a) = \mathbb{E}[\text{succ}|Z = z, A = a]$. Then $s_O^*(O, a) = s_Z^*(Z, a)$ almost surely for ρ -a.e. a , and hence

$$\text{ess sup}_a s_O^*(O, a) = \text{ess sup}_a s_Z^*(Z, a) \quad \text{almost surely.}$$

On a fixed finite candidate set in which every action has positive mass, the ordinary maxima coincide almost surely.

Proof: Definition 1, together with the encoder Markov chain, gives $p(y|O, Z, a) = p(y|Z, a) = p(y|O, a)$ almost surely for ρ -a.e. a . The success functionals therefore agree and share the same essential supremum. Equality of ordinary maxima does not follow in general, because two functionals agreeing ρ -a.e. may differ on a ρ -null set that contains the pointwise optimizer; it does follow on any finite set whose actions each carry positive mass, which is the deployment case (A3). ■

Theorem 2 (η is minimal sufficient): The posterior-predictive field $\eta(o)$ of (7) is a minimal sufficient statistic of o for the action-conditioned outcome family.

Proof: $\eta(o)$ determines $p(y|o, a)$ by construction, so it is sufficient. For any sufficient t , Definition 1 gives $p(y|o, a) = p(y|t(o), a)$, so the map $a \mapsto p(y|o, a)$ is a measurable function of $t(o)$, i.e., $\eta = F \circ t$. A statistic that is a measurable function of every sufficient statistic is minimal [35]. ■

Theorem 3 (The constrained ideal recovers minimal sufficiency): Consider the ideal constrained program

$$\min_q I(Z; O) \quad \text{subject to} \quad I(Z; Y|A) = I(O; Y|A). \quad (17)$$

Assume the constraint is attainable and a minimizer exists. Every feasible encoder is sufficient, and any minimizer has the smallest mutual-information rate among sufficient encoders. For deterministic encoders, its induced partition equals the η -partition up to null sets.

Proof: By Definition 1 and the Markov chain, the equality constraint is exactly sufficiency. The objective then minimizes $I(Z; O)$ over that feasible class. For deterministic statistics, Theorem 2 makes η a function of every feasible statistic, and the coarsest feasible partition is therefore the η -partition modulo null sets. ■

Remark 1 (The trained system is an approximation of Theorem 3, not an instance): The implemented objective differs from the constrained ideal on every count: it optimizes the variational bounds, not the exact informations; it fixes $p_0(z) = \mathcal{N}(0, I_d)$ rather than optimizing the prior, so the rate bound (10) is loose by $\text{KL}(q_\theta(z)||p_0)$; the encoder is a finite-dimensional diagonal Gaussian that need not represent $\eta(o)$; and, as Section IV makes explicit, training minimizes the marginal log-loss of the mixture predictor $\bar{p}_\psi(y|o, a) = \mathbb{E}_q[p_\psi(y|z, a)]$, not the conditional cross-entropy $\mathbb{E}_q[-\log p_\psi]$ that the variational relevance bound (11) contains. Theorem 3 is therefore the motivating ideal; the reported models optimize a rate-regularized marginal objective and have no proved minimal-sufficiency or value-preservation guarantee. The Lagrangian (9) approaches the constrained interpretation only under additional existence and convergence conditions not asserted here.

Theorem 4 (Compression has bounded value cost): Let Z be ϵ -sufficient, $I(O; Y|A) - I(Z; Y|A) \leq \epsilon$, and let A be independent of (O, Z) on a fixed finite candidate set of size m with $\rho(a) \geq \rho_{\min} > 0$. Then

$$\mathbb{E}\left[\max_a s_O^*(O, a)\right] - \mathbb{E}\left[\max_a s_Z^*(Z, a)\right] \leq \sqrt{\frac{m\epsilon}{2\rho_{\min}}} \quad (18)$$

where s_O^* and s_Z^* are the true success functionals of Theorem 1.

Proof sketch: Under (5), write the information deficit as $\sum_a \rho(a) D_a$, where $D_a = \mathbb{E}[\text{KL}(p(y|O, a)||p(y|Z, a))]$. Pinsker's inequality [31] and Jensen's inequality give $\mathbb{E}|s_O^*(O, a) - s_Z^*(Z, a)| \leq \sqrt{D_a/2}$. Using $\max_a u_a - \max_a v_a \leq \sum_a |u_a - v_a|$, Cauchy-Schwarz, and $\sum_a D_a \leq \epsilon/\rho_{\min}$ yields (18). ■

Theorem 4 bounds value loss in terms of the true outcome-information deficit ϵ . It does not bound value directly in terms of removed bits or the empirical KL rate surrogate. The frontier of Section VIII-E therefore reports an empirical association rather than a test of (18).

V. IDENTIFIABILITY: WHAT OUTCOMES CANNOT TELL

The analysis uses an explicit local chart. Rotations use the logarithm chart $\|r\| < \pi$ with a fixed branch, so the axis-angle coordinate is unique and smooth on a neighborhood of x and its Jacobian is well defined; the state coordinates (1) carry the Euclidean metric they are written in, which fixes the inner product used for orthogonal complements below. For a *finite* probe set $\{a_j\}_{j=1}^N$ define the probe map $\Phi_N(x) = (\vartheta_j(x))_{j \leq N} \in \mathbb{R}^{Np}$, each $M(\cdot|x, a_j)$ represented by a finite sufficient parameter vector $\vartheta_j(x) \in \mathbb{R}^p$ (success logit, margin and slip moments), with differential

$$J_N(x) : T_x \mathcal{X} \rightarrow \mathbb{R}^{Np}, \quad J_N(x) \delta = (D_x \vartheta_j(x)[\delta])_{j \leq N}. \quad (19)$$

Φ_N is a finite diagnostic of the selected probe family. Equality on a dense probe sequence can determine a continuous outcome field, but convergence of the derivative kernels requires additional joint derivative-continuity and rank-stability assumptions not imposed here. The experiments use only J_N at the finite N evaluated.

Theorem 5 (Local identifiability, finite-probe form): In the chart above, with Φ_N of locally constant rank (A4) and C^1 (A6), the level set $\Phi_N^{-1}(\Phi_N(x))$ is, in a neighborhood of x , a smooth submanifold with

$$T_x \Phi_N^{-1}(\Phi_N(x)) = \ker J_N(x), \quad \dim = d_{\mathcal{X}} - \text{rank } J_N(x). \quad (20)$$

Any readout $g(x)$ is locally constrained by this finite probe family only along row $J_N(x)$; its identifiable component is the projection of dg onto $(\ker J_N(x))^\perp$, orthogonal in the fixed coordinate metric.

Proof: Locally the level set is $\Phi_N^{-1}(\Phi_N(x))$. Under constant rank (A4) the constant-rank theorem [36] makes it a submanifold with tangent space $\ker D_x \Phi_N(x) = \ker J_N(x)$; the finite codomain $\mathbb{R}^{Np}$ makes the theorem directly applicable without a sequence-space topology. Directional derivatives of g along $\ker J_N(x)$ are unconstrained by Φ_N , giving the split. Rank is read at a stated singular-value tolerance. ■

Proposition 2 (Symmetry orbits lie in the class): Let a group G act on $\mathcal{X}$ and let

$$H = \{g \in G : \Phi(g \cdot x') = \Phi(x') \text{ for every } x' \in \mathcal{X}\}$$

be a subgroup of globally outcome-preserving transformations. Then $H \cdot x \subseteq [x]$ for every x : each image on that orbit is manipulation-indistinguishable from x . A representation of a state known only through the outcome family should therefore not distinguish points on a nontrivial H -orbit.

Proof: Closure of H under composition and inverse follows from global outcome preservation. For $g \in H$, $\Phi(g \cdot x) = \Phi(x)$, so $g \cdot x \sim x$ by Definition 3. ■

Remark 2 (No global decomposition is claimed): No equality is asserted between $[x]$ and the union of local leaves over an outcome-preserving orbit. That would require every connected component of $[x]$ to be joined to the orbit by an outcome-preserving path, a transitivity condition the invariance alone does not supply. The proposition establishes only the stated orbit inclusion; whether a chosen latent family should be multimodal depends on how state uncertainty is represented and is not implied by the discriminative training objective used here.

Corollary 1 (Full-state metrics can penalize probe-null directions): Any pose or shape error metric that penalizes perturbations along $\ker J_N(x)$ assigns error to infinitesimal state changes that the selected probe family cannot distinguish. Standard metrics such as ADD-S and Chamfer distance [17] may do so when their own symmetry treatment does not quotient the same directions; exact object symmetries can instead receive zero set-based error. For this finite diagnostic, the locally constrained component is the row $J_N(x)$ projection, and rankings induced by a full-state metric need not agree with rankings by realized outcome.

Remark 3 (Scope of the theorem): When $J_N(x)$ is rank-deficient, equation (20) identifies continuous finite-probe ambiguity directions. When $\text{rank } J_N = d_{\mathcal{X}}$, it instead says that the local level set is zero-dimensional and rules out such continuous directions; it does not rule out isolated global ambiguities. On our 13-object corpus full finite-probe rank occurs for 6 objects, while cylinders and bottles carry null

spaces of dimension 1–3 (Section VIII-D). The result is local and conditional on the chosen probes, coordinate metric, and rank tolerance.

When a state-space readout is required for interoperability, one possible regularized construction is a distribution over a compact admissible state set. Let P_0 be a proper reference prior, let $\vartheta_N(x) = (\vartheta_j(x))_{j \leq N}$, and let $\hat{\vartheta}_\psi(z)$ express the head predictions for the same probes in the same finite parameterization. Define

$$\begin{aligned} \hat{Q}_z &= \arg \min_{Q \ll P_0} \text{KL}(Q \| P_0) \\ \text{s.t. } &\left\| \mathbb{E}_{x \sim Q} [\vartheta_N(x)] - \hat{\vartheta}_\psi(z) \right\| \leq \delta. \end{aligned} \quad (21)$$

The common finite parameterization makes the constraint typed, while the proper prior and compact domain prevent the unbounded differential-entropy problem. Existence still depends on feasibility and standard lower-semicontinuity conditions; equation (21) is a proposed readout, not an evaluated component of the system.

VI. INFERENCE: DECIDE, FILTER, SENSE, ADAPT

All deployment behavior derives from the two trained modules. Nothing in this section introduces a new learned component except the optional amortized acquisition regressor of (31), which estimates a functional of the existing two.

A. Decision Rule

Given o , draw K encoder samples $z_1, \dots, z_K \sim q_\theta(\cdot | o)$ and define, for each candidate action a ,

$$\begin{aligned} s(o, a) &= \mathbb{E}_{z \sim q} [\varsigma(f_\psi(z, a))] \\ &\approx \frac{1}{K} \sum_k \varsigma(f_\psi(z_k, a)), \end{aligned} \quad (22)$$

$$\begin{aligned} v(o, a) &= \text{Var}_{z \sim q} [\varsigma(f_\psi(z, a))] \\ &\approx \frac{1}{K} \sum_k (\varsigma(f_\psi(z_k, a)) - s)^2. \end{aligned} \quad (23)$$

s marginalizes the spread of the stochastic encoder q_θ ; v is the variance that spread induces on the prediction. Here q_θ is a stochastic encoder distribution and v is an epistemic-style spread, without claiming it is a calibrated Bayesian posterior: the training objective does not identify its variance as epistemic, and conformal calibration concerns outcome prediction sets, not latent coverage. The learned quantity s in (22) is distinct from the true functionals s_O^*, s_Z^* in Theorems 1 and 4. Under realizability and exact predictive fitting it estimates s_O^* . The estimator in (23) is the maximum-likelihood variance (divisor K): because v is compared against the absolute threshold τ_U in (32) and scaled by the absolute λ in (24), Bessel inflation by $K/(K-1)$, 14% at $K=8$, is not a reparameterization but a change in how often the system decides to look again. Selection is risk-averse over a finite candidate set $\mathcal{A}_{\text{cand}} \sim \rho$:

$$a^*(o) = \arg \max_{a \in \mathcal{A}_{\text{filt}}(o)} [s(o, a) - \lambda v(o, a)], \quad \lambda \geq 0, \quad (24)$$

where the argmax runs over the singleton-filtered set defined next. The ordering is filter first and select second; Theorem 6 does not by itself cover the selected extreme.

B. Pairwise Conformal Filtering and Abstention

For the distribution-free construction, the exchangeability unit is a scene. From each of n independent calibration scenes, select one action index by a prespecified randomization independent of its outcomes, giving a held-out fold $\{(o_i, a_i, \text{succ}_i)\}_{i=1}^n$ disjoint from training. Compute nonconformity scores

$$E_i = \begin{cases} 1 - s(o_i, a_i), & \text{succ}_i = 1, \\ s(o_i, a_i), & \text{succ}_i = 0, \end{cases} \quad (25)$$

and let $\hat{q}$ be the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, taken as an order statistic, not an interpolated quantile: interpolation returns a value strictly below the required rank whenever the level falls between ranks and silently under-covers. The construction is non-vacuous only when $n \geq \lceil 1/\alpha - 1 \rceil$; below that no finite $\hat{q}$ retains a singleton and the implementation refuses rather than clamps. The prediction set at a test pair and the singleton-filtered action set are

$$C(o, a) = \{\ell \in \{0, 1\} : \text{score}_\ell(o, a) \leq \hat{q}\}, \quad (26)$$

$$\mathcal{A}_{\text{filt}}(o) = \{a \in \mathcal{A}_{\text{cand}} : C(o, a) = \{1\}\}, \quad (27)$$

with $\text{score}_1 = 1 - s$ and $\text{score}_0 = s$. If $\mathcal{A}_{\text{filt}}(o) = \emptyset$ the system abstains. The singleton test in (27) is essential: checking only $1 - s \leq \hat{q}$ admits any action whose set is the ambiguous $\{0, 1\}$, and at a realistic $\hat{q} \approx 0.68$ that admits the entire band $s \in [0.32, 0.68]$, including a coin flip. The singleton rule (27) instead fails closed. The term ‘‘coverage guarantee’’ is reserved for the exchangeable random-pair unit of Theorem 6, not for the subsequently selected action.

Theorem 6 (Marginal pairwise coverage): Let each calibration pair be obtained from an independent scene by the same outcome-independent action randomization, and let the test pair be generated likewise from a fresh scene. Then $\mathbb{P}(\text{succ} \in C(o, a)) \geq 1 - \alpha$.

Proof: Split-conformal validity [24]: the score is computed with a model fit on data disjoint from the calibration fold, so calibration and test scores are exchangeable, the test rank is uniform, and the quantile bound follows. ■

Remark 4 (What Theorem 6 does and does not certify): Two gaps separate this guarantee from a statement about the executed grasp. First, the exchangeable unit is a random (o, a) pair, but Algorithm 2 executes $a^* = \arg \max_{a \in \mathcal{A}_{\text{filt}}(o)} [s - \lambda v]$: selecting the extreme of a singleton-filtered set is not distributed like a random pair, and selection among N_a candidates can inflate false singletons. Second, the scanned-corpus diagnostic reported in Section VIII-C used all eight actions from each of approximately 2,000 scenes. Those 16,000 individual scores are clustered and do not satisfy the theorem’s exchangeability premise; changing the standard error or informally replacing n by the scene count does not repair that quantile. The scanned-corpus numbers are therefore descriptive empirical pairwise coverage only. Recalibration with one prespecified action per independent scene is required before applying the theorem there. A scene-level calibration of the whole candidate-generation-and-selection pipeline, or a max-over-candidates nonconformity score, is the route to a proved selected-action guarantee.

Under nonexchangeable deployment drift, the following adaptive-level recursion is evaluated, motivated by adaptive conformal inference [29]:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad (28)$$

where α is the *fixed* target, $\text{err}_t \in \{0, 1\}$ indicates miscoverage at step t , and $\gamma > 0$ is a step size.

Proposition 3 (Anchoring with a bounded level sequence): If the unprojected recursion (28) remains bounded, then

$$\frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha = \frac{\alpha_1 - \alpha_{T+1}}{\gamma T} \rightarrow 0.$$

Practical ACI projects the level into a compact interval and requires the corresponding projected-update analysis [29]. Replacing the fixed target by the mutating level produces the open-loop coefficient $1 + \gamma > 1$ for exogenous errors and is therefore not a stable substitute; in our full-feedback stress test it collapsed a 90% target to 1% coverage within roughly one hundred steps.

The identity does not analyze projection, selective feedback, or local coverage. The deployment gap is feedback availability: Algorithm 2 obtains err_t only after an executed grasp yields a label: abstained scenes and unexecuted candidates supply none, so the true feedback is selective and delayed. The experiments update along an offline stream of all realized labels (Table VI); this is a full-feedback stress test of the recursion, not the selective-feedback process the deployed policy would see. No selective-feedback coverage claim is made.

C. One Outcome Model for Sensing and Adaptation

Define the outcome-space ambiguity

$$U(o) = \mathbb{E}_{a \sim p}[v(o, a)] \approx \frac{1}{|\mathcal{A}_{\text{cand}}|} \sum_a v(o, a), \quad (29)$$

a per-scene scalar, never reduced over a batch, because sensing is a per-scene decision. Viewpoint selection explicitly optimizes U ; adaptation uses the same outcome model but a different variational objective.

1) Active viewpoint selection: For a sensing action $b \in \mathcal{B}$ with next-observation predictive $\tilde{p}(o_b|o, b)$, the one-step value of information is

$$\text{IG}(b) = U(o) - \mathbb{E}_{o_b \sim \tilde{p}}[U(o \cup o_b)], \quad b^* = \arg \max_b \text{IG}(b), \quad (30)$$

where $o \cup o_b$ is the representation re-encoded from the fused view set. Negative gain is meaningful and must not be clipped: fusing a poor view can worsen the representation's predictive spread. Computing (30) exactly requires a generative observation model, which this framework exists to avoid; in simulation the true state is known, so o_b is rendered and IG_{true} evaluated exactly. An amortized regressor $\alpha_\omega(o, b)$ is then trained by

$$\mathcal{L}_{\text{acq}}(\omega) = \mathbb{E}[(\alpha_\omega(o, b) - \text{IG}_{\text{true}}(x, o, b))^2], \quad (31)$$

and at deployment a single forward pass replaces the look-ahead. Sensing triggers only when

$$U(o) > \tau_U \quad (32)$$

and budget remains. Fusion is performed *inside* the encoder by a permutation-invariant set aggregation over per-view features. The alternative, encoding views independently and fusing the Gaussians by precision, assumes conditional independence that camera views of one object do not have; measured on our corpus it awards the largest gain to re-fusing the current view with itself, i.e., to not moving, because the arithmetic double-counts the same evidence.

2) Test-time adaptation: Executing a probe a_p and observing y_p defines the generalized Bayesian latent target

$$q'(z) \propto q_\theta(z|o) p_\psi(y_p|z, a_p), \quad (33)$$

approximated variationally: with $q' = \mathcal{N}(\mu', \text{diag } \sigma'^2)$ initialized at $q_\theta(\cdot|o)$,

$$\min_{\mu', \sigma'} \mathbb{E}_{z \sim q'}[-\log p_\psi(y_p|z, a_p)] + \text{KL}(q' \| q_\theta(\cdot|o)), \quad (34)$$

optimized for a few gradient steps with a hard trust region $\|\mu' - \mu\| \leq \delta$. For a scene-perturbing probe with known transition, the prior in (34) is re-encoded from a fresh observation o' and the update is otherwise identical.

Proposition 4 (Fixed point of the variational update): The stationary point of $\min_{q'} \mathbb{E}_{q'}[-\log p] + \lambda \text{KL}(q' \| q_0)$ is $q^* \propto q_0 p^{1/\lambda}$. Hence (34) recovers the latent update (33) iff $\lambda = 1$. Choosing $\lambda = 0.1$ to temper the update instead produces $q_0 p^{10}$, the single-probe likelihood raised to the tenth power. The temperature parameter is therefore not a trust region; the trust region is the hard bound on $\|\mu' - \mu\|$. This characterizes the optimizer over *unrestricted* distributions; the implementation restricts q' to a diagonal Gaussian and takes a few projected steps, so at $\lambda = 1$ it yields a projection of (33) onto the Gaussian family, not the unrestricted update.

Proof: Setting the first variation of $\mathbb{E}_{q'}[-\log p] + \lambda \mathbb{E}_{q'}[\log q'/q_0]$ over all distributions to zero under the normalization constraint gives $\log q^* = \log q_0 + \frac{1}{\lambda} \log p + \text{const}$; restricting to a parametric family replaces the exact stationary point by its information projection. ■

Equation (33) is exact for the auxiliary latent model that treats $q_\theta(z|o)$ as a prior and $p_\psi(y_p|z, a_p)$ as a likelihood. The training objective does not establish that q_θ is a calibrated posterior over physical state, so this is a generalized or pseudo-Bayesian adaptation rule rather than a claim of exact physical Bayesian inference.

Sensing (30) and adaptation (34) share the same outcome model $\tilde{p}_\psi$. Viewpoint selection maximizes the expected reduction of $U(o)$, whereas (34) minimizes a variational free energy and a realized probe can even raise the posterior-predictive variance. Thus the two procedures reuse one outcome model but optimize different functionals. The gradient of (34) must still flow through sampled z , not only the encoder mean; evaluating the head at μ makes $\partial \text{NLL} / \partial \log \sigma^2 \equiv 0$, so the variance cannot move. This probe update is not evaluated on the scanned-object corpus; it is exercised only on the synthetic oracle, and wiring it into closed-loop touch is future work.

D. Algorithms and Complexity

Algorithm 1 is the training loop; Algorithm 2 is the deployment loop. Both mirror the released implementation line for line.

Input: scene prior $p(x)$; sensor $p(o|x)$; target action measure ρ ; proposal ν ; simulator kernel M_{sim} ; analytic margin m_{box} ; budget β ; samples K ; epochs E , anneal length E_R

Output: $\theta, \psi, \hat{q}$

```

1: initialize  $\theta, \psi$ ;  $p_0(z) \leftarrow \mathcal{N}(0, I_d)$ 
2: for epoch  $e = 1$  to  $E$  do
3:    $w_R \leftarrow \min(1, e/E_R)$  rate anneal
4:   for each minibatch do
5:     sample  $x^{(b)} \sim p(x)$ ; render  $o^{(b)} \sim p(o|x^{(b)})$ 
     ( $B, V, 4, H, W$ )
6:     sample  $a^{(b,i)} \sim \nu(\cdot|x^{(b)})$ ; set  $w^{(b,i)} \propto d\rho/d\nu(\cdot|x^{(b)})$  and
     normalize in the batch ( $B, N_a, 7$ )
7:     query (succ, slip)  $\sim M_{\text{sim}}(\cdot|x^{(b)}, a^{(b,i)})$  and set margin =
      $m_{\text{box}}(x^{(b)}, a^{(b,i)})$ 
8:      $(\mu, \log \sigma^2) \leftarrow \text{Enc}_\theta(o)$ ;  $z_k = \mu + \sigma \odot \epsilon_k$ ,  $k \leq K$  ( $B, K, d$ )
9:      $\hat{y} \leftarrow \text{Head}_\psi(z, a)$  over the full  $K \times N_a$  cross product
     ( $B, K, N_a, 6$ )
10:     $D_j \leftarrow$  weighted  $\hat{D}^{\text{marg}}$  of (16) per outcome dim.  $j$ 
11:     $R \leftarrow$  mean of (13) over the batch
12:     $\mathcal{L} \leftarrow \sum_j \beta_j D_j + w_R R$ ; step  $(\theta, \psi)$  by  $\nabla \mathcal{L}$  Eq. (12)
13:  end for
14: end for
15: (optional) train  $\alpha_\omega$  by (31) with rendered look-ahead
16: for formal split-conformal calibration, preselect one action
    independently per calibration scene and compute  $\hat{q}$  by (25)–(26)

```

Algorithm 1: Training the outcome-bottleneck representation

Input: observation o ; $\theta, \psi, \omega, \hat{q}$; K, N_a, λ, τ_U , sensing budget B_{sense} ; ACI state $(\alpha, \gamma, \alpha_t)$

Output: a pairwise-filtered action, or ABSTAIN

```

1:  $(\mu, \log \sigma^2) \leftarrow \text{Enc}_\theta(o)$ 
2: loop
3:    $z_1, \dots, z_K \sim \mathcal{N}(\mu, \text{diag } e^{\log \sigma^2})$ ; sample  $\mathcal{A}_{\text{cand}} \sim \rho$ ,
    $|\mathcal{A}_{\text{cand}}| = N_a$ 
4:   compute  $s[a], v[a]$  by (22)–(23);  $U \leftarrow \text{mean}_a v[a]$ 
5:   if  $U > \tau_U$  and  $B_{\text{sense}} > 0$  then active sensing, Eq. (30)
6:      $b^* \leftarrow \arg \max_b \alpha_\omega(o, b)$ ; execute  $b^*$ ; fuse views;
      $B_{\text{sense}} \leftarrow B_{\text{sense}} - 1$ 
7:      $(\mu, \log \sigma^2) \leftarrow \text{Enc}_\theta(o \cup o_{b^*})$ ; continue
8:   end if
9:    $\mathcal{A}_{\text{filt}} \leftarrow \{a : 1 - s[a] \leq \hat{q} \wedge s[a] > \hat{q}\}$  singleton test,
   Eq. (27)
10:  if  $\mathcal{A}_{\text{filt}} = \emptyset$  then return ABSTAIN
11:   $a^* \leftarrow \arg \max_{a \in \mathcal{A}_{\text{filt}}} (s[a] - \lambda v[a])$ ; execute  $a^*$ ; observe  $y_p$ 
12:  optional full-feedback level update:  $\alpha_t \leftarrow \alpha_t + \gamma(\alpha - \text{err}_t)$ ;
   refresh  $\hat{q}$  at level  $\alpha_t$  Eq. (28)
13:  if task continues then update  $(\mu, \log \sigma^2)$  by (34) and persist
   it for the next decision
14:  return  $a^*, y_p$ 
15: end loop

```

Algorithm 2: Deployment: perceive, filter, act, sense, adapt, or abstain

1) *Complexity and measured interface cost:* Per decision, the sensor node runs one encoder pass, $O(F)$ with F the backbone cost, and transmits $2d = 128$ scalars; the decision node evaluates the head over the $K \times N_a$ cross product, $O(KN_a h)$ with $h = 256$. Table II reports the budget measured on the released checkpoint with fp32 PyTorch on two x86-64 CPU cores (median of 60 runs after 10 warmup iterations): the full perceive-and-score cycle costs 16.2 ms, a 62 Hz decision rate

TABLE II
MEASURED INTERFACE BUDGET (OURS; RELEASED $\alpha=0.1$ CHECKPOINT, FP32, TWO X86-64 CPU CORES, MEDIAN OF 60 RUNS). NO GPU OR ACCELERATOR IS USED.

Quantity	Value
Encoder parameters	11.41 M
Head parameters	0.251 M
Encoder payload ($\mu, \log \sigma^2$), fp32 / fp16	512 B / 256 B
Single RGB-D frame, fp32	147.5 kB
Eight-view set, fp32	1.18 MB
Encoder forward, 1 / 2 threads	25.6 / 14.9 ms
Head, $K=32 \times N_a=8$, 1 / 2 threads	2.1 / 1.3 ms
Perceive-and-score cycle, 1 / 2 threads	27.7 / 16.2 ms

with no GPU and no vendor accelerator, and the transmitted encoder parameters occupy 512 B against 147.5 kB for one raw frame.

Optional sensing adds one amortized forward pass per candidate viewpoint; optional adaptation adds $O(k)$ head evaluations for k gradient steps, with the encoder untouched. There is no meshing, no reconstruction, no dynamics rollout, and no reinforcement-learning loop anywhere in training or deployment. Against a reconstruct-then-plan stack the asymptotic saving is structural: the geometry that pipeline must estimate, store, and ship does not exist here, and the entire perception-to-decision traffic is the encoder parameter pair.

VII. EXPERIMENTAL SETUP

A. A Two-Tier Physics Stack that Measures the Reconstruction Gap

Validating the thesis requires measuring the gap between the geometry a perception pipeline recovers and the physics that happens, with everything else held fixed. Two distinct maps are defined over the *same* states and actions:

- the **analytic tier** produces the deterministic margin $m_{\text{box}}(x, a)$ on an oriented bounding box fitted to the object, approximately the fidelity a pose-and-shape pipeline delivers, and scores each grasp by a differentiable Ferrari–Canny ϵ -quality [11]: contacts on the box surface, linearized friction cones with 8 generators, wrenches $w = [f; (p-c) \times f] \in \mathbb{R}^6$ with torques scaled by a characteristic length, and ϵ evaluated as the inradius of the wrench hull via its support function on 128 fixed directions plus the 12 canonical axes;
- the **simulation tier** defines $M_{\text{sim}}(\text{succ}, \text{slip}|x, a)$ by rolling the identical grasp out in MuJoCo [37] against the object’s true convex decomposition (roughly 20 hulls per object, shipped with the asset), with the scene’s sampled mass, friction, and center of mass: jaws close for 300 steps at 2 ms, contacts settle for 100, and the object is lifted 0.12 m at 0.15 m/s and held for 400 steps; success requires clearing 0.05 m, and slip is the realized drop during the hold.

For the scanned-corpus training loss, the generic task kernel M of Section III-B is instantiated as the hybrid law

$$M_{\text{train}}(dy|x, a) = M_{\text{sim}}(dsucc, dslip|x, a) \cdot \delta_{m_{\text{box}}(x, a)}(dmargin). \quad (35)$$

Thus success and slip are simulated physical outcomes, while the deliberately downweighted margin coordinate is an analytic auxiliary label. The identifiability experiment states separately whether it differentiates the synthetic task kernel or the analytic surrogate. On a parametric box the two tiers see the same geometry and the gap is identically zero, which is why boxes cannot test the claim; on a scanned ketchup bottle the bounding box is a self-consistent but incorrect surface, which exposes the reconstruction gap. The two tiers disagree even on simple geometry: over 320 matched grasps the analytic tier declares success at rate 0.68 against the simulator’s 0.25, agreement is 0.42, the analytic false-positive rate is 0.74, and the quasi-static slip estimate is uncorrelated with dynamic slip ($r = -0.02$). Training the encoder on the analytic tier alone would therefore teach it to be confidently wrong. The dominant success and slip supervision therefore comes from the simulation tier, while the analytic margin remains a low-weight auxiliary label and the same analytic score serves as the *baseline*. The evaluated reconstruct-then-plan pipeline uses an oriented bounding box scored by one differentiable Ferrari–Canny approximation; it is representative of, but does not stand in for, every geometry-certifying pipeline, and the conclusions below are stated against this baseline on this simulator and candidate set rather than against geometric perception in general.

The object set is the 13 scanned grocery objects shipped with the LIBERO benchmark [38]: 6 with a single-box collision hull (butter, cookies, cream cheese, chocolate pudding, popcorn, BBQ sauce) and 7 curved (cans, bottles, cartons). Scenes are generated by dropping an object with randomized yaw and position onto a plane and letting it settle; friction is log-normal with $\log \mu_f \sim \mathcal{N}(\log 0.8, 0.3^2)$, mass $\log m \sim \mathcal{N}(\log 0.25, 0.5^2)$, and the center of mass is displaced inside the hull, so scenes that render identically grasp differently. Each scene carries 8 candidate grasps from a boundary-focused proposal and an 8-camera ring of 96×96 RGB-D views. Fold sizes are 8,000/1,000/2,000/2,000 scenes (train/validation/calibration/test); the calibration and test folds contribute 16,000 grasp-outcome pairs each, of which 11,979 test grasps are kinematically executable and enter predictor evaluation. Calibration and test folds are drawn i.i.d. from the same scene prior with disjoint seeds. Thus A5 holds when one outcome-independent action is preselected per scene as in Theorem 6; it does not hold for the flattened all-pair score array used by the scanned-corpus diagnostic. Deployment drift is evaluated only as the full-feedback stress test associated with Proposition 3. The realized success rate is 0.597, with per-object base rates spanning 0.043 to 0.999, a spread whose consequences for metric choice Section VII-D addresses.

The encoder is a ResNet-18 [39] with a 4-channel stem

(ImageNet-initialized RGB filters, depth initialized from the mean RGB filter), global pooling to 256 features, and a linear head to $(\mu, \log \sigma^2) \in \mathbb{R}^{128}$, $d = 64$; view sets are fused by a permutation-invariant mean over per-view features. The outcome head embeds the 7-D action to 256 units, modulates the embedding by feature-wise linear transformation conditioned on z so that the action-pose product enters structurally rather than being discovered, and emits the six parameters of (14). Training uses AdamW (learning rate 5×10^{-4} , weight decay 10^{-4}), batch size 128, 60 epochs, bf16 autocast with the loss in fp32, gradient clipping at 1.0, $K = 8$ encoder samples through the marginal estimator of Proposition 1, a 15-epoch rate anneal, and a 10-epoch deterministic warmup; simulator and solver settings are pinned in configuration and logged with every run. Unless stated otherwise $\beta_{\text{succ}} = 300$, $\beta_m = \beta_s = 10^{-2}$; the pairwise-filter experiments use $\beta_{\text{succ}} = 30$, and the frontier sweeps $\beta_{\text{succ}} \in \{1, 3, 10, 30, 100\}$. Every number in Section VIII is written by the released experiment scripts into JSON and typeset from there; the implementation carries 97 unit and regression tests, including named regressions for each defect analyzed in Sections IV–VI.

C. Synthetic Validation Oracle

The diagnostic is first evaluated where the analytic answer is known. The kernel M is instantiated as a closed-form synthetic operator over a 6-D state with vector observations, constructed so that its finite-probe indistinguishability class, and hence $\ker J_N(x)$ (rank $J_N = 3$ everywhere), is known analytically. An MLP encoder ($d = 16$, $\beta_{\text{succ}} = 20$) and the identical head, calibration, and inference stack run end to end on it in seconds, which pins pipeline defects before any conclusion is drawn from the simulator; its 64,000-point test fold also sharpens the coverage estimates of Section VIII-C.

D. Evaluation Protocol

1) *Metrics*: Predictive validity is AUC against the realized lift outcome. Deployment value is the success rate of *executed* grasps as a function of the act rate (the fraction of scenes the system commits to), plus top-1 success when always acting. Pairwise-filter quality is empirical marginal coverage $\mathbb{P}(\text{succ} \in C(o, a))$ against the target $1 - \alpha$, singleton precision $\mathbb{P}(\text{succ}=1 | a \in \mathcal{A}_{\text{flt}})$, abstention rate, and retained-action fraction; these are distinct estimands and are reported separately. Theorem 6 applies only when calibration and test each use one prespecified random action per independent scene. The flattened scanned-corpus all-pair numbers do not satisfy that premise and are labeled descriptive. Representation cost is the rate R in nats and the interface size in scalars.

2) *Pooled versus within-scene AUC*: With base rates spanning 0.043 to 0.999, a pooled AUC rewards a model for recognizing *which object* is present rather than *which grasp* will hold: a predictor carrying object identity and nothing else already pools to ≈ 0.72 here. Accordingly, both pooled and within-scene AUC are reported. Within-scene AUC ranks only the candidate grasps of one settled scene and is averaged over scenes containing at least one success and one failure, so object recognition alone cannot improve it. The two metrics

can order models oppositely (a Simpson effect we observed in an early head whose pooled score exceeded both of its strata), and ablation conclusions in Section VIII-H rest on the within-scene metric exclusively.

3) *Baselines and privileged state*: The analytic Ferrari–Canny score on the fitted bounding box is the reconstruct-then-plan surrogate, given the same candidate actions as ours. Random selection and the majority class calibrate the floor. An MLP trained on the *true* 14-D physical state (1), an input no perception system can access, is a privileged baseline: its within-scene AUC is 0.685. Because the 14-vector omits mesh identity and detailed shape, it is not an upper bound on observation-based prediction. Learned point-cloud grasp networks [8], [10] are not included because their action samplers, gripper models, and training corpora differ from ours in ways that confound the single question this evaluation isolates, namely what the *estimand* of perception should be under a fixed candidate set and controller. Section IX returns to this scope decision.

4) *Statistical reporting*: Actions sharing a scene are clustered, so action-level binomial and AUC standard errors are not used. The reported tables give point estimates from the designated primary checkpoint unless marked otherwise; valid uncertainty requires a scene-clustered bootstrap or an equivalent cluster-respecting analysis, which is not available in the present artifact. Distortions are reported with the standard error over test batches. The full model is retrained end to end with three independent seeds: within-scene AUC 0.635 ± 0.011 , pooled AUC 0.875 ± 0.002 , top-1 success 0.692 ± 0.008 , selective precision 0.976 ± 0.008 (mean $\pm$ sd), with descriptive all-pair coverage 0.900–0.901 on every retraining. Ablation variants are trained once; their deltas are interpreted against this measured noise floor without significance claims below it.

VIII. RESULTS

A. E1: Does Reconstructed-Geometry Quality Predict the Lift?

Hypothesis. If the reconstruct-then-plan estimand were sound, a wrench-space quality computed on the reconstruction would rank grasps by their realized outcome. The sufficiency thesis predicts instead that the score is informative only where the reconstruction happens to be correct, and uninformative or worse elsewhere.

Data. Table III: the analytic proxy attains 0.542 pooled and 0.548 within-scene AUC, against 0.876 and 0.639 for ours; the privileged-state baseline reaches 0.685 within-scene. The representation thus recovers most of what a model with direct access to pose, size, friction, mass and center of mass achieves, from pixels alone, while the proxy stays near chance. No percentage of a ceiling is reported because the 14-vector is not a sufficient state for the corpus (it lacks shape and identity) and so is not a valid upper bound. No threshold on the proxy yields an accuracy meaningfully above the majority class (0.612 vs. 0.597). These results establish little ranking value for the realized outcome; they do not by themselves assess probability calibration.

The control group. Six of the thirteen objects *are* boxes: for them the bounding-box reconstruction is exact and no surface

TABLE III

PREDICTIVE VALIDITY AGAINST THE REALIZED LIFT OUTCOME. 11,979 EXECUTABLE GRASPS ON 13 SCANNED OBJECTS; BASE SUCCESS RATE 0.597. THE PRIVILEGED-STATE ROW IS AN MLP ON THE 14-VECTOR (1), AN INPUT UNAVAILABLE TO ANY PERCEPTION SYSTEM. BECAUSE THAT VECTOR OMITTS OBJECT IDENTITY AND MESH, IT DOES NOT BY ITSELF DETERMINE THE SIMULATOR OUTCOME ACROSS THE CORPUS, SO IT IS A BASELINE, NOT A TRUE UPPER BOUND; RESIDUAL ROLLOUT STOCHASTICITY BOUNDS ALL PREDICTORS BELOW 1.

Predictor	Pooled	Within-scene	Best acc.
Chance / majority	0.500	0.500	0.597
Ferrari-Canny (recon.)	0.542	0.548	0.612
Ours	0.876	0.639	0.784
<i>Privileged state</i>	—	0.685	—

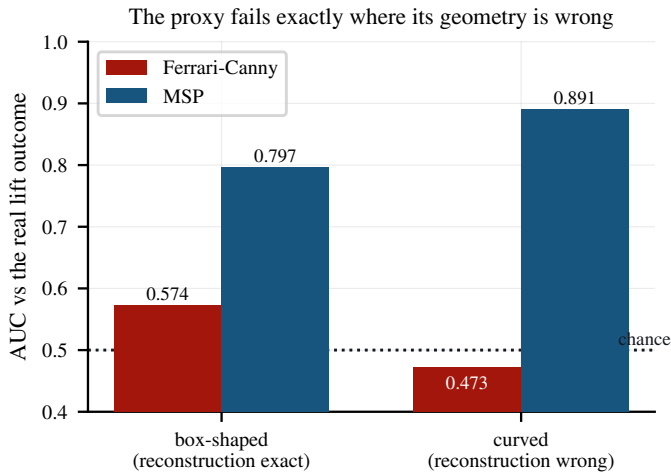


Fig. 2. Pooled AUC against the realized lift, stratified by whether the bounding-box reconstruction is exact (box-hulled objects) or wrong (curved objects). The analytic score degrades to below chance where its geometry is hallucinated; the outcome-trained representation improves there.

is hallucinated, so the thesis predicts the proxy should work there if it works anywhere. Fig. 2 confirms the prediction in both directions: on box-hulled objects the proxy reaches 0.574, weak but above chance, while on curved objects it falls to 0.473, below chance, meaning the grasps it prefers are systematically the grasps that fail. Ours moves the opposite way, 0.797 on boxes and 0.891 on curved objects, because curvature is visible in RGB-D and informative about outcomes even when no box can represent it. On a parametric-box benchmark this experiment is uninformative because reconstruction and truth coincide. Scanned meshes are therefore necessary to expose the reconstruction gap.

A cell that ships geometry and certifies force closure on it inherits a score whose errors concentrate on the objects where geometry is hardest, and no link bandwidth or reconstruction compute repairs an estimand anti-correlated with the outcome on curved stock, at least for this bounding-box reconstruction and this analytic metric; a stronger learned reconstruction under the same controller is the control this result invites next.

B. E2: Selective Execution

Hypothesis. If a confidence score tracks realized outcomes, executed success should increase as the act rate decreases. A

TABLE IV

DEPLOYMENT VALUE: THE SYSTEM RANKS A SCENE’S CANDIDATES, EXECUTES ITS FAVORITE, AND COMMITS ONLY ON ITS MOST CONFIDENT SCENES. ENTRIES ARE SUCCESS RATES OF EXECUTED GRASPS OVER 1,997 TEST SCENES.

Act rate	Ferrari-Canny	Ours
0.25 (selective)	0.503	0.984
1.00 (always act)	0.665	0.692
Random grasp	0.623	

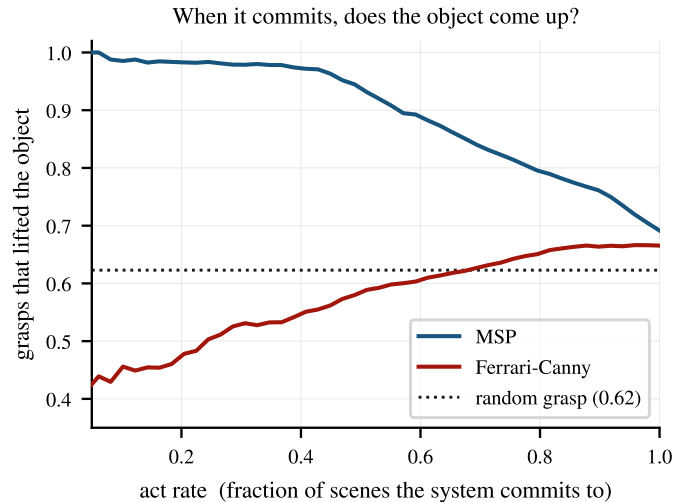


Fig. 3. Risk-coverage behavior. Success of executed grasps as the act rate sweeps from selective to always-on. The proxy’s curve falls as it grows more selective, crossing below the random-grasp floor; ours rises to 0.984 at a 25% act rate.

reversed trend indicates that selectivity concentrates failures.

For each method, the scene confidence is the score of its own selected top-ranked action: $s - \lambda v$ for ours and the Ferrari-Canny value for the analytic baseline. The 25% operating point retains the top quartile of scenes under that method-specific score. The table reports point estimates; a scene-clustered confidence interval is not available in the present artifact.

Data. Table IV. Always acting, both systems sit near the observed top-1 range (0.665 proxy, 0.692 ours, 0.623 random). The regimes separate under selectivity: at a 25% act rate the proxy’s executed grasps succeed at 0.503, below picking a grasp at random, while ours reach 0.984. The proxy’s confidence ordering is inverted on the curved half of the corpus (Fig. 2), so conditioning on its confidence selects for failure.

Deployment implication. Abstention, deferral, and handoff rely on a confidence ordering that improves outcomes as the act rate decreases. At the reported 25% operating point, the learned score reaches 0.984 success while the analytic score falls below random selection.

C. E3: Pairwise Filtering and Coverage Diagnostics

Hypothesis. Theorem 6 promises marginal coverage for one outcome-independently selected action from each independent scene. The synthetic oracle uses independent test points and directly checks that setting. The scanned-corpus experiment predates the corrected scene-level calibration protocol and

TABLE V

EMPIRICAL SINGLETON FILTERING ON THE SCANNED-OBJECT CORPUS ($\beta_{\text{succ}} = 30$; 16,000 ACTION RECORDS OVER $\approx 2,000$ SCENES). BECAUSE CALIBRATION POOLED EIGHT CORRELATED ACTIONS PER SCENE, THESE VALUES ARE DESCRIPTIVE AND THEOREM 6 DOES NOT APPLY. SINGLETON PRECISION IS A DISTINCT SELECTION-CONDITIONAL ESTIMAND.

	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$
Target coverage $1 - \alpha$	0.950	0.900	0.800
Pairwise coverage (Ours)	0.946	0.897	0.800
Singleton precision	0.792	0.744	0.673
Abstention rate	0.823	0.743	0.598
Retained action fraction	0.169	0.235	0.373

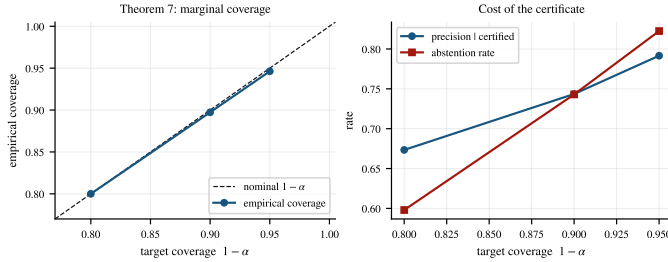


Fig. 4. Empirical against target coverage on the synthetic oracle (64,000 test points per level, where the outcome kernel is the analytic map with a closed-form indistinguishability class). The split-conformal sets track the diagonal at every level; the right panel relates the result to the KL rate surrogate.

pools all eight actions per scene. It is retained as a clustered empirical diagnostic, not as a distribution-free validation.

Data. Table V gives descriptive all-pair coverage 0.946/0.897/0.800 against targets 0.950/0.900/0.800. These close point estimates do not establish finite-sample validity because calibration and test scores are clustered by scene. On the independently sampled synthetic oracle (Section VII-C), the construction gives 0.979/0.949/0.901/0.802/0.700 against targets 0.98/0.95/0.90/0.80/0.70. In the scanned diagnostic, singleton precision is observed to rise with the target coverage level; this monotonicity is empirical, not required by conformal validity. At $\alpha = 0.1$ the filter retains 23.5% of candidate actions and abstains on 74.3% of scenes. Singleton precision is selection-conditional and is not controlled by Theorem 6.

On the abstention rate. A photograph does not reveal friction, mass, or center of mass, and those variables influence slip and torque failure. The 74% empirical abstention rate is consistent with substantial unresolved ambiguity, but the clustered experiment is not a 90% action-level safety certificate. Active touch (Section VI-C) is intended to reduce this ambiguity rather than executing anyway on an analytic score whose false-positive rate is 0.74 (Section VII-A). Raising the rate lowers the price: at $\beta_{\text{succ}} = 100$ the same nominal filter level gives abstention 0.367 and singleton precision 0.848 (Table VIII).

Empirical filtering under stream drift. Deployment violates exchangeability. The anchored recursion is stress-tested on the released checkpoint by streaming the 16,000 test decisions sorted by object identity into thirteen homogeneous segments whose base rates sweep 0.043 to 0.999, a severe local shift.

TABLE VI

EMPIRICAL COVERAGE UNDER AN OBJECT-SORTED STREAM (OURS; 16,000 DECISIONS SORTED BY OBJECT, TARGET 0.90), A FULL-FEEDBACK STRESS TEST: EVERY DECISION SUPPLIES A LABEL, UNLIKE THE SELECTIVE FEEDBACK OF A DEPLOYED POLICY THAT ABSTAINS. THE FIXED COLUMN USES THE CALIBRATION-FOLD QUANTILE; THE ACI COLUMN USES (28) WITH $\gamma = 0.01$. MARGINAL COVERAGE IS ORDER-INVARIANT; LOCAL COVERAGE IS NOT.

	Fixed $\hat{q}$	ACI (anchored)
Marginal coverage	0.899	0.900
Worst 500-decision window	0.702	0.744
Worst object segment	0.720	0.834
Per-object dispersion (std)	0.111	0.035

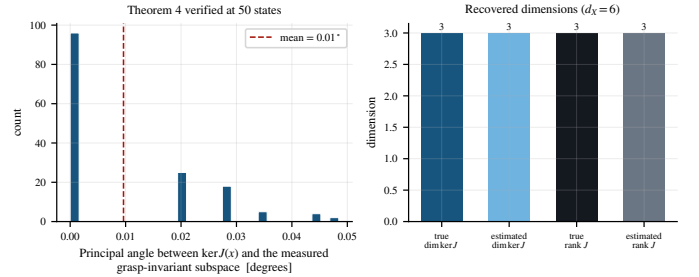


Fig. 5. Verification of Theorem 5 on the synthetic oracle, whose indistinguishability class is known in closed form: principal angles between the predicted $\ker J_N(x)$ and the measured invariant subspace concentrate near zero (mean 0.013° , max 0.056° over 50 states), and the recovered rank and null dimensions match the ground truth exactly.

This is a *full-feedback* test: every decision supplies a label, which the deployed policy would not have on abstained scenes, so it bounds the best case of selective feedback rather than reproducing it. Reordering does not change the empirical all-pair marginal average, which remains 0.90; what the fixed filter loses is *local* coverage, collapsing to 0.702 over the worst 500-decision window and 0.720 on the worst object segment. The anchored recursion at $\gamma = 0.01$ recovers 4.2 points of worst-window and 11.5 points of worst-object coverage and compresses per-object dispersion from 0.111 to 0.035, at identical marginal coverage. It does not fully restore the nominal level inside the worst window because adaptation lags each segment boundary by $O(1/\gamma)$ decisions. The reported values retain this shortfall.

D. E4: Identifiability, Predicted vs. Measured

Hypothesis. Theorem 5 predicts that the local level-set tangent space for the selected probes equals $\ker J_N(x)$. Two maps must be kept distinct. On the *synthetic oracle*, M is the analytic map, so differentiating Φ_N and comparing to the closed-form $\ker J_N$ is a self-consistent verification of the diagnostic and of the theorem for that M . On the *scanned objects*, the analytic bounding-box tier is differentiated, whereas the physical outcome kernel is the simulator on the true mesh; the two are different maps, so the scanned-object numbers test whether the analytic $\ker J_N$ predicts the simulator’s locally invariant directions, not the theorem’s J_N for the true physics. Both experiments are reported with their distinct interpretations.

TABLE VII

FINITE-PROBE IDENTIFIABILITY PER OBJECT. $\dim \ker J_N$ IS THE DIMENSION OF THE ANALYTIC PROBE-MAP LEVEL-SET TANGENT SPACE AT THE SETTLED STATE; THE PRINCIPAL ANGLE COMPARES THE ANALYTIC-TIER $\ker J_N(x)$ AGAINST THE SIMULATOR'S INDEPENDENTLY MEASURED OUTCOME-INVARIANT SUBSPACE ON THE TRUE MESH. THIS IS A SURROGATE-AGREEMENT MEASUREMENT, NOT A COMPUTATION OF THE THEOREM'S J_N FOR THE SIMULATOR KERNEL (SEE TEXT). FOR 6 OF 13 OBJECTS J_N HAS FULL RANK AND THE LOCAL LEVEL SET IS ZERO-DIMENSIONAL (REMARK 3). THE COMPARISON USES THE FIXED PROBE SET, PERTURBATION SCALE, AND RANK TOLERANCE OF THE COMPANION DIAGNOSTIC CONFIGURATION. THOSE NUMERICAL SETTINGS ARE NOT INCLUDED IN THE PRESENT MANUSCRIPT ARTIFACT, SO REPRODUCING THE ANGLES REQUIRES THAT CONFIGURATION.

Object	rank J_N	$\dim \ker J_N$	Max angle [°]
alphabet soup	12	2	0.05
BBQ sauce	14	0	—
butter	14	0	—
chocolate pudding	14	0	—
cookies	14	0	—
cream cheese	14	0	—
ketchup	13	1	0.05
macaroni and cheese	13	1	0.04
milk	11	3	0.21
orange juice	12	2	0.55
popcorn	14	0	—
salad dressing	12	2	0.06
tomato sauce	12	2	0.06

Data. On the synthetic oracle of Section VII-C, where M is the analytic map and the theorem applies exactly, the predicted and measured subspaces agree to a mean principal angle of 0.013° and a maximum of 0.056° over 50 states, with rank and null dimension recovered exactly (Fig. 5). This verifies the theorem and the diagnostic. On the scanned corpus (Table VII), the analytic-tier $\ker J_N$ agrees with the simulator's measured invariant subspace to below 0.56° on every rank-deficient object, and the null directions are the physically expected ones, rotation about a can's symmetry axis and combinations the gripper never loads. This supports the analytic surrogate as a predictor of the simulator's local invariances on these objects, not as a verification that residual ambiguity equals $\ker J_N$ for the true physics, which would require differentiating an emulator of the simulator validated against held-out rollouts. The six box-hulled objects have full-rank J_N : the analytic finite-probe level set is locally zero-dimensional, though isolated or unprobed ambiguities may remain.

Systemic implication. The analytic J_N is computable before deployment and can flag local directions that the chosen surrogate probe map does not resolve. On scanned objects, agreement with measured simulator invariances is empirical and does not prove that a vision-only endpoint can never recover those coordinates. Corollary 1 motivates reporting the finite-probe row/null decomposition alongside full-state pose metrics.

E. E5: The Rate-Distortion Frontier

Hypothesis. Raising β_{succ} in the implemented objective should emphasize success log-loss at the price of a larger KL rate surrogate. This is an empirical optimization hypothesis:

TABLE VIII

RATE-DISTORTION FRONTIER OVER THE SUFFICIENCY BUDGET (OURS; 16,000 TEST POINTS PER ROW). β MULTIPLIES THE SUCCESS RELEVANCE TERM, SO LARGER β EMPHASIZES SUCCESS PREDICTION; THE UNWEIGHTED TOTAL $\sum_j D_j$ IS DOMINATED BY THE DELIBERATELY SACRIFICED MARGIN AND SLIP LIKELIHOODS AND NEED NOT FALL. DESCRIPTIVE ALL-PAIR COVERAGE REMAINS NEAR 0.90 ALONG THE FRONTIER, WITHOUT A FINITE-SAMPLE CLAIM FOR THE CLUSTERED CALIBRATION.

β	R [nats]	D_{succ}	$\sum_j D_j$	Coverage	Abstention
1	0.004	0.656	0.725	0.896	0.643
3	0.088	0.622	0.767	0.892	0.674
10	0.280	0.598	0.870	0.904	0.736
30	0.534	0.582	0.993	0.897	0.743
100	3.286	0.412	0.956	0.905	0.367

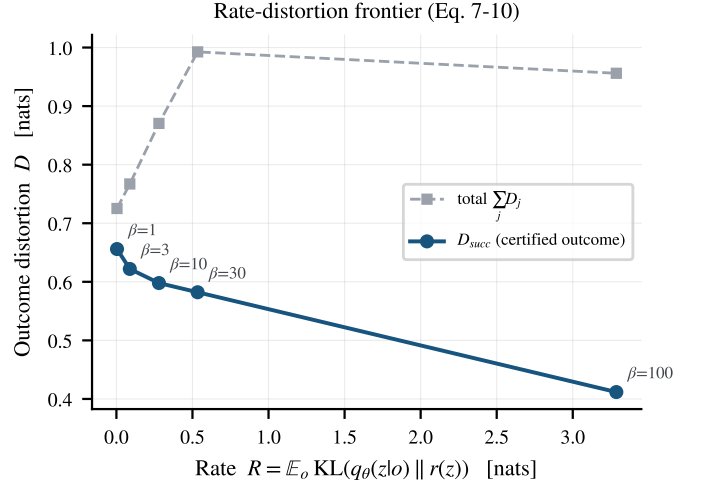


Fig. 6. The measured rate-distortion frontier (12). Each point is one trained model; the abscissa is the KL rate surrogate R (Section III), not a coded wire length, and the ordinate is outcome distortion.

Theorem 3 concerns the exact constrained ideal, and Theorem 4 concerns an unmeasured true information deficit.

Data. Table VIII. The frontier shows the intended empirical tradeoff: D_{succ} falls monotonically, 0.656 to 0.412 nats, as the rate rises from 0.004 to 3.29 nats. The descriptive all-pair coverage stays between 0.892 and 0.905 across the operating range, but the operating price changes: the extra 2.75 nats between $\beta = 30$ and $\beta = 100$ halve the abstention rate (0.743 to 0.367) and lift singleton precision from 0.744 to 0.848. In this experiment, the rate surrogate is therefore associated with how often the filter retains an action; it is not a proved coding-rate or coverage law.

Interface accounting. Two quantities must be separated. The *operational payload* is what the sensor node transmits: 128 floating-point parameters, 512 B in fp32 or 256 B in fp16, and this is a measured, deployed number. The *KL rate surrogate* R is the regularizer of (12), an upper bound on $I(Z; O)$; it is 0.53 nats at $\beta=30$ and 3.29 nats at $\beta=100$. It can motivate a relative-entropy coding design against a shared prior [32], but no such coder, its quantization, or its finite-block overhead is implemented or timed here. The value R is not treated as an achieved or finite-block code length; the demonstrated operational reduction is the 512 B payload against 147.5 kB for

one frame (288 $\times$) and 1.18 MB for the 8-view set (2304 $\times$).

F. E6: Active Perception with Independent Scoring

Hypothesis. Acquiring the viewpoint that maximizes (30) should reduce ambiguity by more than a random second view; and because the selector and the scorer share Monte-Carlo noise, selecting and scoring on the same estimate must inflate the apparent gain (a winner’s curse), so an unbiased evaluation selects on one estimate and scores on an independent one.

Data. A second view of any kind reduces ambiguity $U(o)$ by 13.8%; choosing it by (30), scoring with an independent Monte-Carlo estimate, buys 16.7%. Reusing the selection draw produces a downward-biased estimate of fused ambiguity, so that statistic is discarded. The independent estimate is the reported result. Its modest size shows that most of the value of a second view is in taking one at all, and the residual ambiguity after both views is dominated by unobservable physics that no viewpoint resolves, the regime where probing (Section VI-C) rather than looking is the intended next acquisition. These are point estimates from the reported run; replicated Monte Carlo or scene-level uncertainty is not available in the present artifact.

G. E7: Combined Deterministic and No-Rate Ablation

Hypothesis. A deterministic point-estimate system provides a capacity-matched comparison to the stochastic encoder. The implemented variant uses the same backbone, head, optimizer, and schedule, but sets $z = \mu$ and removes the KL term. It is therefore a combined ablation of sampling and rate regularization, not an isolation of either mechanism.

Data. The point-estimate baseline (within-scene AUC 0.527, pooled 0.705, top-1 0.633, selective precision 0.651, against 0.639/0.876/0.692/0.984 for the full model) sits at chance on the metric that matters: its action-response standard deviation is 0.0008 against 0.221 for the full model, meaning it emits one number per scene and ignores the action outright; its selective precision at a 25% act rate is 0.651, barely above executing random grasps (0.623), against 0.984 for the full model. Its pooled AUC of 0.705 sits at the recognition-only level, the signature of the collapse of Section IV: under the combined change, the head predicts the per-object base rate. Its descriptive all-pair coverage is 0.897 at a nominal 0.90 filter level, but it cannot rank actions sharply. The experiment shows that the full training configuration matters; matched variants with sampling and KL regularization changed separately are required to attribute the gain to either component.

H. E8: Ablations

Hypothesis. Each architectural commitment of Sections IV–VI was made against a measured failure; removing it should reproduce that failure. All comparisons use the within-scene AUC (Section VII-D); *action response* is the standard deviation of $s(o, a)$ across a scene’s candidates, whose collapse toward zero means the model has stopped ranking grasps and emits one number per scene.

Data. Table IX. (i) *The budget.* Replacing the per-dimension budget with a uniform β hands capacity out in proportion

TABLE IX
ABLATIONS ON THE SCANNED-OBJECT CORPUS ($\beta_{\text{succ}} = 300$ EXCEPT WHERE THE BUDGET ITSELF IS ABLATED). THE POOLED AUC IS REPORTED TO EXPOSE ITS FAILURE AS A REFEREE: IT SCORES THE UNIFORM-BUDGET AND NO-PERCEPTION VARIANTS NEAR 0.7 ALTHOUGH BOTH HAVE STOPPED RANKING GRASPS.

Variant	Within	Pooled	Resp.
Ours (full)	0.639	0.875	0.221
Uniform outcome budget	0.511	0.710	0.002
Constant z	0.506	0.615	0.031
$K=1$ sample	0.602	0.844	0.174
$d=16$	0.634	0.873	0.219
$d=32$	0.587	0.846	0.185

to the accidental magnitude of each likelihood; the Bernoulli success term, the only outcome the decision rule reads, is the smallest of the three, and the model stops attending to the action: action response collapses from 0.221 to 0.002 and the within-scene AUC falls to 0.511, chance. On this corpus the margin coordinate is the Ferrari–Canny value computed on the bounding box, so a uniform budget spends the representation’s capacity fitting precisely the proxy this paper shows is uninformative. (ii) *Perception.* Replacing the representation with a constant (the head can still exploit action priors) gives 0.506 within-scene, confirming the head does not bypass z ; on the synthetic oracle the same ablation moves the *total* outcome distortion $\sum_j D_j$ from -0.730 to $+0.656$ nats, a 1.386-nat gap. (The total can be negative because the continuous margin and slip terms are Gaussian log-densities; the Bernoulli success term D_{succ} alone is non-negative, and it rises by 0.28 nats under the same ablation.) (iii) *Marginalization.* One encoder sample costs 3.7 within-scene points (0.602 vs. 0.639), the measured value of multi-sample marginalization at selection time; it is not by itself evidence that the spread is calibrated epistemic uncertainty. (iv) *Latent width.* $d = 16$ matches $d = 64$ (0.634 vs. 0.639) at a quarter of the interface, consistent with a low intrinsic dimension of the manipulation-relevant variation; $d = 32$ scores lower (0.587) under a single seed. Measured seed noise on the full configuration is ± 0.011 within-scene, so the budget and constant-representation ablations (13-point drops) exceed the full-model seed noise by more than tenfold, though the ablation variants are single-seed so their own variance is not measured; the $K=1$ drop (3.7 points) is comparable to a few full-model standard deviations, and the latent-width differences are at the noise floor. These data do not establish latent-width saturation. (v) *Abstention.* On the independent synthetic oracle, acting only on singleton sets raises success-when-acting from 0.935 to 0.949 while abstaining on 9 of 400 episodes. The effect is small in this synthetic setting. The separate scanned-corpus selective experiment shows a larger difference between methods, but it is not an action-level conformal guarantee.

Pooled AUC as a failed referee. The pooled column would have passed variants (i) and (ii) at 0.710 and 0.615 although both stopped discriminating between a scene’s grasps; every headline conclusion of this paper would survive the substitution of pooled for within-scene AUC, but the ablation

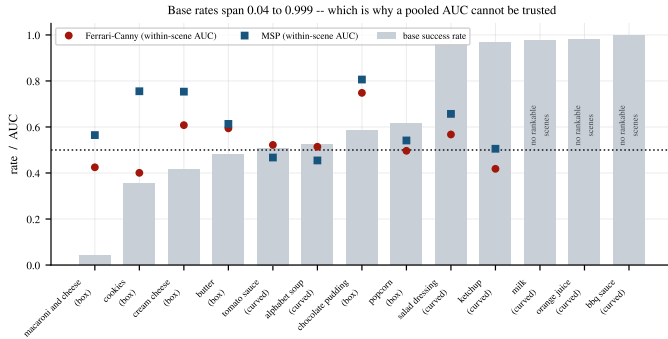


Fig. 7. Per-object base success rates (0.043 to 0.999) with pooled and within-scene AUC. The spread is why a pooled AUC can be won by object recognition alone, and why single-class or near-single-class scenes make within-scene AUC undefined or high-variance for the extreme-base-rate objects (Section VIII-I).

TABLE X

LEAVE-ONE-OBJECT-OUT TRANSFER (OURS). EACH ROW IS A MODEL TRAINED WITH THE NAMED OBJECT ABSENT AND EVALUATED ONLY ON THAT OBJECT’S TEST SCENES; THE LEGACY ALL-PAIR FILTER IS CALIBRATED ON THE OTHER TWELVE OBJECTS (NOMINAL TARGET $1 - \alpha = 0.90$). WITHIN-SCENE AUC IS AVERAGED ONLY OVER SCENES CONTAINING BOTH LABELS. BBQ SAUCE HAS TOO FEW VALID SCENES FOR A STABLE ESTIMATE AND IS REPORTED AS UNAVAILABLE; THE AGGREGATE WITHIN-SCENE MEAN THEREFORE EXCLUDES THAT ENTRY. TOP-1 SUCCESS AND PAIRWISE COVERAGE REMAIN WELL DEFINED.

Held-out object	base	Within	Top-1	Cov. fix	Cov. ACI
<i>Moderate base rate (0.3 < base < 0.7): within-scene AUC is meaningful</i>					
butter	0.48	0.569	0.474	0.703	0.895
cream cheese	0.41	0.546	0.429	0.543	0.896
tomato sauce	0.51	0.511	0.473	0.785	0.896
alphabet soup	0.52	0.479	0.500	0.774	0.897
choc. pudding	0.58	0.695	0.778	0.845	0.898
popcorn	0.62	0.475	0.601	0.666	0.893
cookies	0.35	0.709	0.491	0.805	0.899
subgroup mean		0.569	0.535	0.732	0.896
<i>Extreme base rate: within-scene AUC degenerate</i>					
macaroni	0.04	0.497	0.028	0.625	0.780
BBQ sauce	1.00	–	0.993	0.752	0.890
salad dress.	0.96	0.520	0.975	0.704	0.877
ketchup	0.97	0.546	0.975	0.533	0.853
milk	0.98	0.528	0.984	0.913	0.907
orange juice	0.98	0.422	0.974	0.820	0.898
subgroup mean		0.503	0.822	0.724	0.868
All objects		0.541	0.667	0.728	0.883

story would silently invert. Metric choice is an architectural decision.

I. E9: Generalization to Unseen Objects

Hypothesis. Every experiment above trains and tests on the same 13 objects, so a within-distribution representation could in principle encode object identity rather than transferable outcome structure. Transfer is tested with 13 models, each trained with one object held out and evaluated on that object. The legacy all-pair filter is fit on the remaining twelve objects and deployed on the held-out one, which provides an empirical object-level shift test for the full-feedback update; it is not covered by Proposition 3 or Theorem 6.

Data. Table X. The results split by base rate. On the seven objects with moderate base rates, where within-scene

ranking is a well-posed question, transfer degrades from the in-distribution 0.639 to 0.569 but stays above chance, with two objects (cookies, chocolate pudding) at 0.70; the representation carries grasp-ranking structure to geometry it has never seen, but less than half of it survives the shift. On the six extreme-base-rate objects, within-scene AUC is near-degenerate: with base rates from 0.96 to 0.999 (and macaroni at 0.043) almost no scene contains both a success and a failure, so the metric rests on very few valid scenes and is not a reliable ranking measure. BBQ sauce is omitted because too few scenes contain both labels; top-1 success is the meaningful quantity in that regime. Excluding that unstable entry, the all-object within-scene mean is 0.541. The representation is partly object-conditional, and the claim that the representation transfers intact across object identity is not supported at this scope; the relevant transfer estimate for the moderate-base-rate subset is 0.569.

Full-feedback adaptation improves empirical pairwise coverage. The reported quantity is a clustered all-pair diagnostic under simulated object-level shift, not the synthetic stream of Section VIII-C. Calibrated on twelve objects and deployed on a thirteenth, the fixed quantile undercovers severely (mean 0.728, worst 0.533 on ketchup, dispersion 0.110), because the held-out object’s score distribution is genuinely out of calibration. The anchored update raises it to 0.883 mean and compresses dispersion to 0.032, at a worst case of 0.780. This is the same mechanism as Table VI exercised with full outcome feedback. The update substantially reduces empirical undercoverage but remains below the 0.90 nominal target on average and establishes neither finite-sample validity nor selected-action or selective-feedback coverage.

IX. DISCUSSION, SCOPE, AND LIMITATIONS

1) *What the evidence licenses:* Within a version-pinned simulated world, the simulator supplies physical success and slip while the analytic tier supplies the downweighted margin label. The experiments support four statements. The reconstruct-then-plan estimand fails where its geometry is wrong, to the point of anti-correlation (E1, E2). The outcome-trained representation ranks grasps near a privileged-state baseline from pixels and sustains a 0.984 selective success rate at the reported operating point (E1, E2). A combined deterministic/no-KL variant collapses to chance-level within-scene ranking, although that comparison does not isolate the responsible mechanism (E7). The independent synthetic oracle tracks the split-conformal targets; the scanned-corpus all-pair coverage is descriptive because of scene clustering (E3, E5). The finite-probe identifiability theorem is verified on the synthetic oracle. On scanned objects, the reported angles measure agreement between an analytic surrogate and simulator invariances (E4).

2) *What it does not license:* No claim transfers to hardware yet. The main missing tier is a low-parameter residual of the outcome head fit on real executed grasps; it is not included. Until it exists, the sufficiency motivation is relative to the simulated task kernel. The generalization limit is explicit (Section VIII-I): the representation as trained is partly object-conditional, its grasp-ranking transfers to unseen objects at

0.569 within-scene AUC against 0.639 in-distribution, and a claim that the sufficiency statistic is object-agnostic is not supported at this scope. Under full outcome feedback, the anchored update improves mean pairwise coverage under simulated object-level shift from 0.728 to 0.883, still below the 0.90 nominal target; finite-sample clustered, selected-action, and selective-feedback guarantees remain open. The encoder distribution is a diagonal Gaussian; richer mixture or flow families could represent multimodal latent uncertainty, but the present discriminative objective does not identify that uncertainty with a physical-state posterior. The encoder pools globally to one code per scene, which suits the evaluated single-object scope but not cluttered multi-object cells. Seed variance of the full model is measured at ± 0.011 within-scene AUC over three retrainings; ablation variants remain single-seed, so cross-architecture deltas below roughly two standard deviations, including the non-monotone latent-width sweep, remain unresolved. External point-cloud grasp networks remain excluded by the controlled protocol (Section VII-D); the internal point-estimate baseline of Section VIII-G is a controlled comparison, but it changes both sampling and KL regularization and therefore does not isolate either effect. A shared-controller comparison against published systems on real hardware, with matched candidate sets and compute, remains necessary for a deployment decision. Finally, the observed active-perception gain is small because much of the residual variation comes from physical parameters not observed by the camera. Closed-loop touch is the next test of whether the probe update in (34) can reduce the abstention rate in Table V.

X. CONCLUSION

This work changes the training target of perception for grasping. Instead of an accurate geometric state, an edge perception node produces a stochastic, outcome-trained representation for the action-conditioned outcome family. An ideal constrained information program characterizes minimal sufficiency; the implemented model optimizes a rate-regularized marginal predictor and transmits a measured 512-byte fp32 payload. The finite-probe theory identifies local level-set directions for a fixed probe family, and split conformal supplies marginal pairwise coverage when one randomized action is taken from each exchangeable scene. Empirically, on 11,979 executed simulated grasps over 13 scanned objects, the standard analytic score on reconstructed geometry predicts lift outcomes at 0.542 AUC and falls below chance on curved stock, while the 128-scalar representation reaches 0.876 and sustains 0.984 success at a 25% commitment rate at 16 ms of CPU compute per decision. Independent synthetic points track the tested conformal levels; the scanned-corpus all-pair coverage remains a clustered empirical diagnostic pending scene-level recalibration. Transfer to unseen objects is partial: moderate-base grasp ranking degrades to 0.569 within-scene AUC, while a full-feedback adaptive update improves mean pairwise coverage from 0.728 to 0.883 under object shift without establishing a selective-feedback guarantee. For networked manipulation, the result motivates shifting the design

question from how accurately geometry can be shipped to how compactly action-relevant outcome information can be represented and empirically validated.

REFERENCES

- [1] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 637–646, 2016.
- [2] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Communications Surveys & Tutorials*, vol. 19, no. 4, pp. 2322–2358, 2017.
- [3] B. Kehoe, S. Patil, P. Abbeel, and K. Goldberg, "A survey of research on cloud robotics and automation," *IEEE Transactions on Automation Science and Engineering*, vol. 12, no. 2, pp. 398–409, 2015.
- [4] B. Wen, W. Yang, J. Kautz, and S. Birchfield, "FoundationPose: Unified 6D pose estimation and tracking of novel objects," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [5] H. Wang, S. Sridhar, J. Huang, J. Valentin, S. Song, and L. J. Guibas, "Normalized object coordinate space for category-level 6D object pose and size estimation," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [6] W. Shen, G. Yang, A. Yu, J. Wong, L. P. Kaelbling, and P. Isola, "Distilled feature fields enable few-shot language-guided manipulation," in *Proc. Conf. Robot Learning (CoRL)*, 2023.
- [7] A. Rashid, S. Sharma, C. M. Kim, J. Kerr, L. Y. Chen, A. Kanazawa, and K. Goldberg, "Language embedded radiance fields for zero-shot task-oriented grasping," in *Proc. Conf. Robot Learning (CoRL)*, 2023.
- [8] H.-S. Fang, C. Wang, M. Gou, and C. Lu, "GraspNet-1Billion: A large-scale benchmark for general object grasping," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [9] H.-S. Fang, C. Wang, H. Fang, M. Gou, J. Liu, H. Yan, W. Liu, Y. Xie, and C. Lu, "AnyGrasp: Robust and efficient grasp perception in spatial and temporal domains," *IEEE Transactions on Robotics*, vol. 39, no. 5, pp. 3929–3945, 2023.
- [10] M. Sundermeyer, A. Mousavian, R. Triebel, and D. Fox, "Contact-GraspNet: Efficient 6-DoF grasp generation in cluttered scenes," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2021.
- [11] C. Ferrari and J. Canny, "Planning optimal grasps," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 1992, pp. 2290–2295.
- [12] M. A. Roa and R. Suárez, "Grasp quality measures: Review and performance," *Autonomous Robots*, vol. 38, no. 1, pp. 65–88, 2015.
- [13] R. Newbury, M. Gu, L. Chumbley, A. Mousavian, C. Eppner, J. Leitner, J. Bohg, A. Morales, T. Asfour, D. Kragic, D. Fox, and A. Cosgun, "Deep learning approaches to grasp synthesis: A review," *IEEE Transactions on Robotics*, vol. 39, no. 5, pp. 3994–4015, 2023.
- [14] E. Calvanese Strinati and S. Barbarossa, "6G networks: Beyond Shannon towards semantic and goal-oriented communications," *Computer Networks*, vol. 190, p. 107930, 2021.
- [15] D. Gündüz, Z. Qin, I. Estella Aguerri, H. S. Dhillon, Z. Yang, A. Yener, K.-K. Wong, and C.-B. Chae, "Beyond transmitting bits: Context, semantics, and task-oriented communications," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 1, pp. 5–41, 2023.
- [16] Y. Xiang, T. Schmidt, V. Narayanan, and D. Fox, "PoseCNN: A convolutional neural network for 6D object pose estimation in cluttered scenes," in *Proc. Robotics: Science and Systems (RSS)*, 2018.
- [17] T. Hodaň, F. Michel, E. Brachmann, W. Kehl, A. G. Buch, D. Kraft, B. Drost, J. Vidal, S. Ihrke, X. Zabulis, C. Sahin, F. Manhardt, F. Tombari, T.-K. Kim, J. Matas, and C. Rother, "BOP: Benchmark for 6D object pose estimation," in *Proc. European Conf. Computer Vision (ECCV)*, 2018.
- [18] J. Mahler, J. Liang, S. Niyaz, M. Laskey, R. Doan, X. Liu, J. A. Ojea, and K. Goldberg, "Dex-Net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics," in *Proc. Robotics: Science and Systems (RSS)*, 2017.
- [19] N. Tishby, F. C. Pereira, and W. Bialek, "The information bottleneck method," in *Proc. 37th Annual Allerton Conf. Communication, Control, and Computing*, 1999, pp. 368–377.
- [20] A. A. Alemi, I. Fischer, J. V. Dillon, and K. Murphy, "Deep variational information bottleneck," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.
- [21] V. Pacelli and A. Majumdar, "Learning task-driven control policies via information bottlenecks," in *Proc. Robotics: Science and Systems (RSS)*, 2020.

- [22] W. Huang, C. Wang, Y. Li, R. Zhang, and L. Fei-Fei, “ReKep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation,” *arXiv preprint arXiv:2409.01652*, 2024.
- [23] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, “OpenVLA: An open-source vision-language-action model,” in *Proc. Conf. Robot Learning (CoRL)*, 2024.
- [24] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.
- [25] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [26] L. Lindemann, M. Cleaveland, G. Shim, and G. J. Pappas, “Safe planning in dynamic environments using conformal prediction,” *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 5116–5123, 2023.
- [27] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh, S. Tu, N. Brown, P. Xu, L. Takayama, F. Xia, J. Varley, Z. Xu, D. Sadigh, A. Zeng, and A. Majumdar, “Robots that ask for help: Uncertainty alignment for large language model planners,” in *Proc. Conf. Robot Learning (CoRL)*, 2023.
- [28] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [29] I. Gibbs and E. Candès, “Adaptive conformal inference under distribution shift,” in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [30] Y. Matsubara, M. Levorato, and F. Restuccia, “Split computing and early exiting for deep learning applications: Survey and research challenges,” *ACM Computing Surveys*, vol. 55, no. 5, pp. 1–30, 2022.
- [31] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ, USA: Wiley, 2006.
- [32] M. Havasi, R. Peharz, and J. M. Hernández-Lobato, “Minimal random code learning: Getting bits back from compressed model parameters,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.
- [33] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2014.
- [34] Y. Burda, R. Grosse, and R. Salakhutdinov, “Importance weighted autoencoders,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2016.
- [35] E. L. Lehmann and G. Casella, *Theory of Point Estimation*, 2nd ed. New York, NY, USA: Springer, 1998.
- [36] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed. New York, NY, USA: Springer, 2012.
- [37] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.
- [38] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” in *Proc. Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.
- [39] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.